Core Publishable Work

From the single-cell birth–death–catastrophe process
to a burst-aware Basic Model of Viral Replication

Summary drawn from Chapters 3–4 and the BMVR extension notes

August 4, 2026

Abstract

This document is a clean, self-contained summary of the results that are ready (or nearly ready) for publication. It has two layers.

1. **Single-cell BDC theory.** Closed forms for survival, moments, burst time, burst size, and the quasi-stationary distribution — with several simplifications, one major correction, and one striking identity that the thesis chapters contain the ingredients for but never state.
2. **Population (BMVR) extension.** A renewal formulation that embeds those single-cell kernels into a within-host model, with exact effective parameters, an invariant basic reproduction number, and a sharp criterion for when bursting beats constant budding at matched R_0 (the "flooding advantage").

Everything stated as a Result has been checked algebraically and numerically. Open technical tasks and thesis errata are deliberately omitted here.

Contents

1 The story in one page

Many within-host models treat infected cells as black boxes that release virions at a constant rate p (the Basic Model of Viral Replication, BMVR). Real lytic pathogens — *Yersinia pestis*, *Francisella tularensis*, lytic phage — do something different: they grow inside a cell and dump their load in a single burst. The question is whether that strategy can be modelled exactly, and what it buys the pathogen early on, when stochastic extinction is still likely.

The route taken here is:

1. model one infected cell as a classical **birth–death–catastrophe (BDC)** process (birth rate β, death μ, load-proportional burst rate δ);
2. derive closed forms for everything that cell does — survival, load, release, burst timing and size;
3. feed those formulas into a **renewal (age-structured) BMVR**, so the population model is no longer phenomenological but is forced by the intracellular process;
4. read off what changes: effective release rate, R_0, growth rate, and especially the probability that the infection dies out in the first generation.

The headline biological result is simple to state:

> At fixed total output per cell, bursting lowers extinction probability relative to constant budding *if and only if*
>
> $$\frac{1}{\delta} < \frac{1}{\beta} + \frac{1}{\mu}.$$
>
> When $\mu = 0$, bursting always wins.

Everything else in this document either supports that claim, or shows how the same single-cell formulas yield effective BMVR parameters and testable single-cell predictions.

2 Notation (used consistently throughout)

Symbol	Meaning	Notes
β, μ, δ	birth, death, catastrophe rates	per capita, intracellular
X_t	intracellular particle count	CTMC
R	absorbing rupture state	
W_t	released count from one cell	0 until burst
τ	burst time	may be ∞
$\mathcal{K}$	burst size (random)	$= X_t$ at burst
$I(t)$	no-burst probability	$\to b$ as $t \to \infty$
$D(t)$	cleared, no burst by t	$= p_0(t)$
$\widehat{I}(t)$	still productively infected	$= I - D$
$J(t), K(t)$	$\mathbb{E}[X_t]$, $\mathbb{E}[X_t^2]$	mean load moments
$V(t)$	$\mathbb{E}[W_t]$	cumulative mean release
a, b	characteristic roots	$b < 1 < a$
A, B	$a-1$, $1-b$	simplify almost everything
θ	$\beta(a-b)$	growth exponent
T	target-cell count (held fixed)	BMVR
γ	infection rate	BMVR (not β)
$\mathcal{I}, \mathcal{V}$	infected cells, free virions	BMVR counts
c	virion clearance	BMVR

3 The single-cell process

3.1 Definition

Starting from $X_t = 1$ (one founder particle), the process jumps

$$n \;\to\; n+1 \quad \text{at rate } \beta n, \qquad n \;\to\; n-1 \quad \text{at rate } \mu n, \qquad n \;\to\; R \quad \text{at rate } \delta n.$$

On the jump to R the cell bursts: W_t is set to the pre-burst load and stays there. The process is classical (Brockwell–Gani–Resnick; Karlin–Tavaré); what matters for applications is the family of closed forms below.

3.2 Root algebra

$$\eta = \frac{\beta+\mu+\delta}{2\beta}, \qquad a = \eta + \sqrt{\eta^2 - \tfrac{\mu}{\beta}}, \qquad b = \eta - \sqrt{\eta^2 - \tfrac{\mu}{\beta}}. \tag{1}$$

Write $A := a-1 > 0$, $B := 1-b > 0$, $\theta := \beta(a-b)$. The identities that do most of the work are

$$ab = \frac{\mu}{\beta}, \qquad AB = \frac{\delta}{\beta}, \qquad A + B = a - b. \tag{2}$$

In particular $b < 1 < a$ always, and $\beta(1-a)(1-b) = -\delta$.

3.3 Three fixation functions

There are three natural absorbing-behaviour probabilities, not two:

$$I(t) = \mathbb{P}\{\text{no burst by } t\}\,, \tag{3}$$

$$D(t) = \mathbb{P}\{\text{internal extinction by } t,\ \text{no burst}\}\,, \tag{4}$$

$$\widehat{I}(t) = \mathbb{P}\{X_t \notin \{0, R\}\} = I(t) - D(t). \tag{5}$$

Both I and D solve the Riccati equation

$$f' = \beta(f-a)(f-b) = \mu - (\beta+\mu+\delta)f + \beta f^2,$$

with $I(0) = 1$ and $D(0) = 0$. Writing $w := \mathrm{e}^{\theta t} \geq 1$:

Result 3.1 (Closed forms for I, D, $\widehat{I}$)**.**

$$I(t) = \frac{aB + bAw}{B + Aw}, \qquad D(t) = \frac{ab\,(w-1)}{aw - b}, \qquad \widehat{I}(t) = \frac{(a-b)^2\,w}{(B+Aw)(aw-b)}. \tag{6}$$

Limits: $I(\infty) = D(\infty) = b$, $\widehat{I}(\infty) = 0$, $\widehat{I}(0) = 1$.

Remark 3.2 (Correction of thesis formula)*.* The formula $\widehat{I}(t) = \hat{\eta}/(1-(1-\hat{\eta})\mathrm{e}^{\hat{\eta}\beta t})$ used in Chapter 3 is wrong whenever $\mu > 0$. It comes from treating $\widehat{I}$ as if it obeyed the same autonomous Riccati as I. The correct branching term on a birth is $I^2 - D^2$, not $\widehat{I}^2$. The error is invisible when $\mu = 0$ (then $D \equiv 0$). For $X_0 = k$ founders one has $\widehat{I}_k = I^k - D^k$, not $\widehat{I}^k$.

3.4 Moments and mean release

Result 3.3 (First and second moments; mean release)**.**

$$J(t) = \mathbb{E}[X_t] = \frac{(a-b)^2\,w}{(B+Aw)^2} = -\frac{\beta}{\delta}(I-a)(I-b), \tag{7}$$

$$K(t) = \mathbb{E}[X_t^2] = \Big[1 + \tfrac{2\beta}{\delta}(1-I)\Big]J, \tag{8}$$

$$V(t) = \mathbb{E}[W_t] = (1-I)\Big[1 + \tfrac{\beta}{\delta}(1-I)\Big]. \tag{9}$$

Two useful simplifications: J no longer carries the cluttered prefactor of the thesis form (the identity $AB = \delta/\beta$ cancels it), and V is a quadratic in $(1-I)$ alone — no separate J or μ.

Result 3.4 (Lifetime-time release and conditional mean burst size)**.**

$$V_\infty = \frac{a(1-b)}{a-1} = \frac{aB}{A}, \qquad \mathbb{E}[\mathcal{K} \mid \mathit{burst}] = \frac{V_\infty}{1-b} = \frac{a}{a-1}. \tag{10}$$

V_∞ includes realisations that clear internally and release nothing; $\mathbb{E}[\mathcal{K} \mid burst]$ conditions them out.

4 Burst time, burst size, and the quasi-stationary distribution

4.1 Burst time

Result 4.1. *$\mathbb{P}\{\tau > t\} = I(t)$, so the (defective) density of burst time is*

$$\varphi(t) = -I'(t) = \delta J(t), \qquad \int_0^\infty \varphi(t)\,\mathrm{d}t = 1 - b.$$

Mass b sits at $\tau = \infty$ (internal clearance without burst).

4.2 State probabilities are geometric at every time

For $n \geq 1$,

$$p_n(t) = \Big(\frac{a-b}{b-aw}\Big)^2 w\ P(t)^{n-1}, \qquad P(t) = \frac{1-w}{b-aw}. \tag{11}$$

Conditional on $X_t \geq 1$, the load is geometric with ratio $P(t)$ at every t, and $P(t) \to 1/a$.

4.3 The identity

Result 4.2 (Burst size = quasi-stationary distribution)**.** *Both are geometric on* $\{1, 2, \dots\}$ *with ratio* $1/a$*:*

$$\mathrm{QSD}(n) = \lim_{t\to\infty} \mathbb{P}\{X_t = n \mid X_t \notin \{0, R\}\} = (a-1)\, a^{-n}, \tag{12}$$

$$\mathbb{P}\{\mathcal{K} = k \mid \textit{burst}\} = (a-1)\, a^{-k}. \tag{13}$$

Unconditionally (including no-burst realisations), $\mathbb{P}\{\mathcal{K} = k\} = (\delta/\beta)\, a^{-k}$ *for* $k \geq 1$*, with mass* b *at* 0*.*

This is the cleanest single-cell result in the project. It is not obvious: bursts are size-biased draws from the transient load, integrated against the burst intensity δn. That the outcome is exactly the QSD deserves a conceptual proof; the algebra is already complete (see below).

Immediate QSD moments:

$$\langle X\rangle_{\mathrm{QS}} = \frac{a}{a-1}, \qquad \mathrm{Var}_{\mathrm{QS}}(X) = \frac{a}{(a-1)^2}. \tag{14}$$

Note $\langle X\rangle_{\mathrm{QS}} = \mathbb{E}[\mathcal{K} \mid \mathrm{burst}]$. When $\mu = 0$ both also equal V_∞; when $\mu > 0$ they differ (e.g. at $(\beta, \mu, \delta) = (1, 0.2, 0.05)$: 17.23 versus 13.99).

4.4 Proof of the burst-size law (no simulation required)

With $P(t) = (1-w)/(b-aw)$ one has

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big[\frac{P^k}{k\beta}\Big] = p_k(t).$$

Hence

$$\mathbb{P}\{\mathcal{K} = k\} = \delta k \int_0^\infty p_k(t)\, \mathrm{d}t = \frac{\delta}{\beta}\big[P^k\big]_0^\infty = \frac{\delta}{\beta}\, a^{-k}.$$

Two independent checks close the loop:

1. $\sum_{k\geq 1} (\delta/\beta) a^{-k} = 1 - b$ (eventual-burst probability);
2. $\sum_{k\geq 1} k(\delta/\beta) a^{-k} = V_\infty$ (mean release).

4.5 Size-biasing and late bursts

A cell bursts at rate δX_t, so given burst at time t the size is size-biased:

$$\mathbb{E}[\mathcal{K} \mid \tau = t] = \frac{K(t)}{J(t)} \xrightarrow[t\to\infty]{} 1 + \frac{2\beta}{\delta}(1-b).$$

A very late burst is roughly twice the size of an average one — the signature of size-biasing a geometric.

5 From one cell to a population: the renewal BMVR

5.1 Why a constant p substitution fails

Chapter 4 attempted to replace the BMVR release rate p by the QSD mean $\langle X \rangle_{\mathrm{QS}}$. That cannot work: p has units of particles per unit time, while $\langle X \rangle_{\mathrm{QS}}$ is a pure count. When $\mu = 0$ one even has $\langle X \rangle_{\mathrm{QS}} = V_\infty$, so the proposal substitutes total lifetime output for an output *rate*.

The deeper issue is that infected cells are not synchronised: at any host time t they have a distribution of ages (times since infection). The right object is therefore an age-structured, or renewal, formulation.

5.2 Classical BMVR (for comparison)

In the simplified (target-cell-constant) Basic Model of Viral Replication one has incidence $i(t) = \gamma T\, \mathcal{V}(t)$ and the ODE system

$$\begin{aligned} \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= \gamma T\, \mathcal{V} - d_{\mathcal{I}}\, \mathcal{I}, \\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= p\mathcal{I} - c\, \mathcal{V}. \end{aligned} \tag{15}$$

Here p is a constant per-cell release rate and $d_{\mathcal{I}}$ a constant infected-cell removal rate. Both are phenomenological: they are not derived from intracellular dynamics.

5.3 The two kernels from the BDC

Both are already closed-form from the single-cell theory:

$$\widehat{I}(a) = \mathbb{P}\{\text{cell of age } a \text{ is still productively infected}\} = \frac{(a-b)^2\, w}{(B+Aw)(aw-b)}, \quad w = \mathrm{e}^{\theta a}, \tag{16}$$

$$g(a) = \delta K(a) = V'(a) = \text{expected release flux at age } a. \tag{17}$$

The release kernel is correct because a cell bursts at rate δX_t releasing X_t particles, so expected release in $[a, a + \mathrm{d}a]$ is $\delta \mathbb{E}[X_t^2]\, \mathrm{d}a = \delta K(a)\, \mathrm{d}a$. Explicitly,

$$K(a) = \left[1 + \tfrac{2\beta}{\delta}(1 - I(a))\right] J(a), \qquad J(a) = \frac{(a-b)^2\, w}{(B+Aw)^2}. \tag{18}$$

5.4 The new viral dynamics equations

With target cells T held constant, infection rate γ, virion clearance c, and incidence

$$i(t) = \gamma T\, \mathcal{V}(t), \tag{19}$$

the burst-aware (renewal) model is the following integro-differential system.

Result 5.1 (New viral dynamics — burst-aware BMVR).

$$\boxed{\begin{aligned} \mathcal{I}(t) &= \mathcal{I}_0\, \widehat{I}(t) + \int_0^t i(t-\alpha)\, \widehat{I}(\alpha)\, \mathrm{d}\alpha, && (20) \\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \mathcal{I}_0\, g(t) + \int_0^t i(t-\alpha)\, g(\alpha)\, \mathrm{d}\alpha - c\, \mathcal{V}(t), && (21) \end{aligned}}$$

with kernels (16) *and* $g = \delta K$*. Equivalently, writing the convolutions more compactly and using* $i = \gamma T\,\mathcal{V}$*,*

$$\boxed{\begin{aligned}\mathcal{I}(t) &= \mathcal{I}_0\,\widehat{I}(t) + \big(\gamma T\,\mathcal{V} * \widehat{I}\big)(t), \qquad (22)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \mathcal{I}_0\,g(t) + \big(\gamma T\,\mathcal{V} * g\big)(t) - c\,\mathcal{V}(t). \qquad (23)\end{aligned}}$$

Here $\mathcal{I}_0$ is the number of cells already infected at $t = 0$ (the initial cohort); their contribution is $\mathcal{I}_0\widehat{I}(t)$ to the infected-cell count and $\mathcal{I}_0 g(t)$ to the release flux. If the infection is seeded only by free virions with $\mathcal{I}_0 = 0$, those terms drop and one recovers the pure convolution form used in the working notes.

What each equation replaces.

Classical BMVR (15)	New system (20)–(21)
$\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$ (constant removal rate $d_{\mathcal{I}}$)	$\mathcal{I} = \mathcal{I}_0\widehat{I} + i * \widehat{I}$ (age-structured survival via $\widehat{I}$)
$\mathcal{V}' = p\mathcal{I} - c\mathcal{V}$ (constant release rate p)	$\mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}$ (age-structured release via $g = \delta K$)

Both constant BMVR terms are replaced: not only release $p\mathcal{I}$, but also cell removal $d_{\mathcal{I}}\mathcal{I}$. Chapter 4's original attempt only ever proposed to change the first.

Why the equations look different. In the classical model, infected cells are memoryless: every cell, regardless of age, contributes the same p and dies at the same $d_{\mathcal{I}}$. In the new model a cell of age α is still productively infected only with probability $\widehat{I}(\alpha)$, and contributes expected release flux $g(\alpha)$. Because cells are infected at different times, the population totals are convolutions of incidence against these kernels — not products of a stock $\mathcal{I}(t)$ with a constant.

Optional differential form for $\mathcal{I}$. Differentiating (20) (fundamental theorem + Leibniz rule) gives

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = i(t) + \mathcal{I}_0\,\widehat{I}'(t) + \int_0^t i(t-\alpha)\,\widehat{I}'(\alpha)\,\mathrm{d}\alpha, \qquad (24)$$

so the incidence $i(t)$ appears as a source term exactly as in the classical model, while the remaining terms encode age-dependent loss of productivity (burst or internal clearance). There is in general no constant $d_{\mathcal{I}}$ for which (24) reduces to $\mathcal{I}' = i - d_{\mathcal{I}}\mathcal{I}$ for all histories of i; such a reduction holds only for pure exponential solutions (Section 7).

Remark 5.2 (What is and is not novel). The renewal/age-structured framework itself is standard (McKendrick–von Foerster; Nelson–Gilchrist–Perelson-type models). The novelty is that the kernels $\widehat{I}$ and g are derived in closed form from a mechanistic intracellular stochastic process, rather than posited phenomenologically — together with the consequences below.

6 Numerical overlays of the two systems

To make the difference between (15) and (20)–(21) concrete, both systems are integrated side by side for a range of intracellular and population parameters.

How to read every plot.

Style	Model
Solid navy line	**NEW model** — renewal / burst-aware BMVR (kernels $\widehat{I}$, $g = \delta$
Dashed orange line with open circles	**CLASSICAL BMVR** — constant release rate p and removal d

Each panel also carries its own legend and the words "NEW" / "CLASSICAL" printed near the curves. The CLASSICAL parameters are the fairest constant-parameter match at the $r = 0$ end of the effective-rate map,

$$p = p_{\text{eff}}(0) = \frac{\beta V_\infty}{\log(a/(a-1))}, \qquad d_{\mathcal{I}} = d_{\mathcal{I},\text{eff}}(0) = \frac{\beta}{\log(a/(a-1))}, \tag{25}$$

so total lifetime output and mean productive lifetime agree. Figures are generated by `figures/plot_bmvr_compa`

6.1 Single-cell kernels (NEW model only)

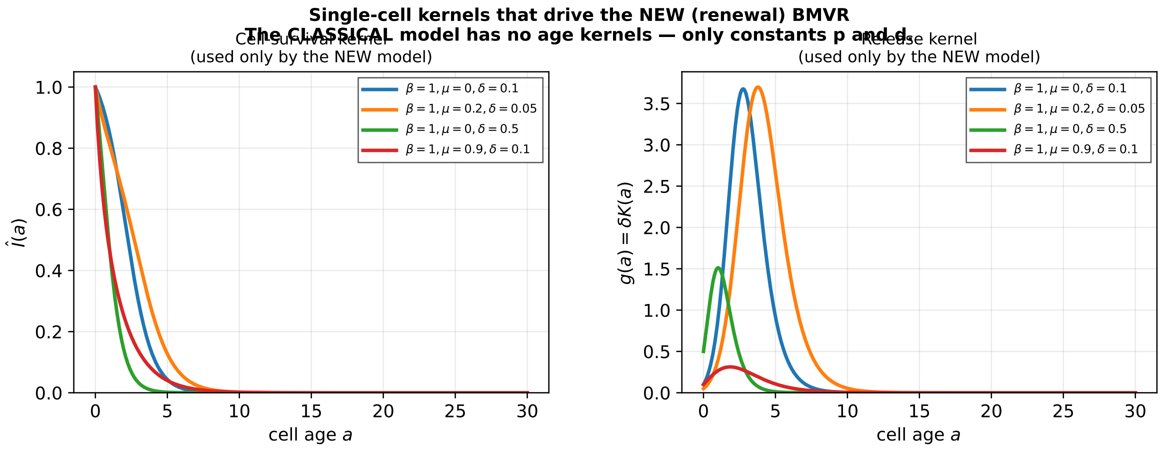

Figure 1: Cell-survival kernel $\widehat{I}(a)$ and release kernel $g(a) = \delta K(a)$ for four intracellular parameter sets. These age-dependent kernels drive the **NEW** model. The **CLASSICAL** model has no kernels — only the constants p and $d_{\mathcal{I}}$.

6.2 Free virions and infected cells

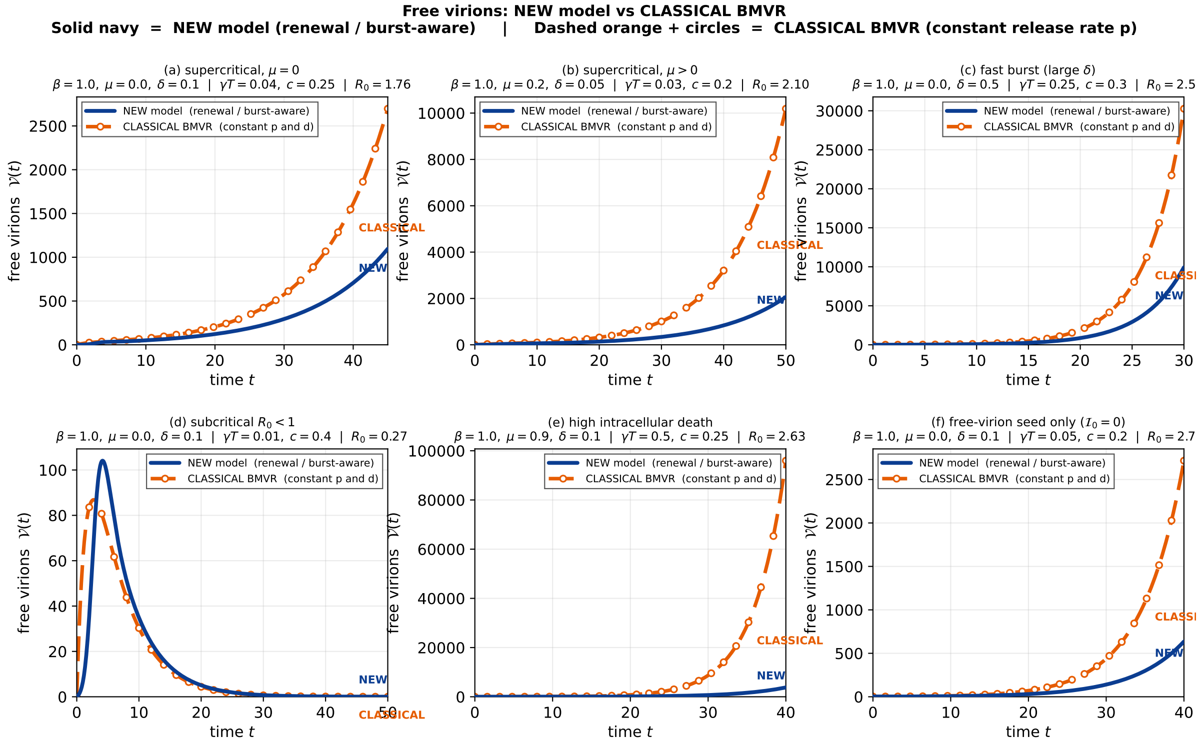

Figure 2: Free-virion trajectories $\mathcal{V}(t)$. **Solid navy = NEW model** (renewal / burst-aware); **dashed orange with circles = CLASSICAL BMVR** (constant p, d). Six scenarios: supercritical growth, subcritical decay, fast burst, high intracellular death, and free-virion-only seeding.

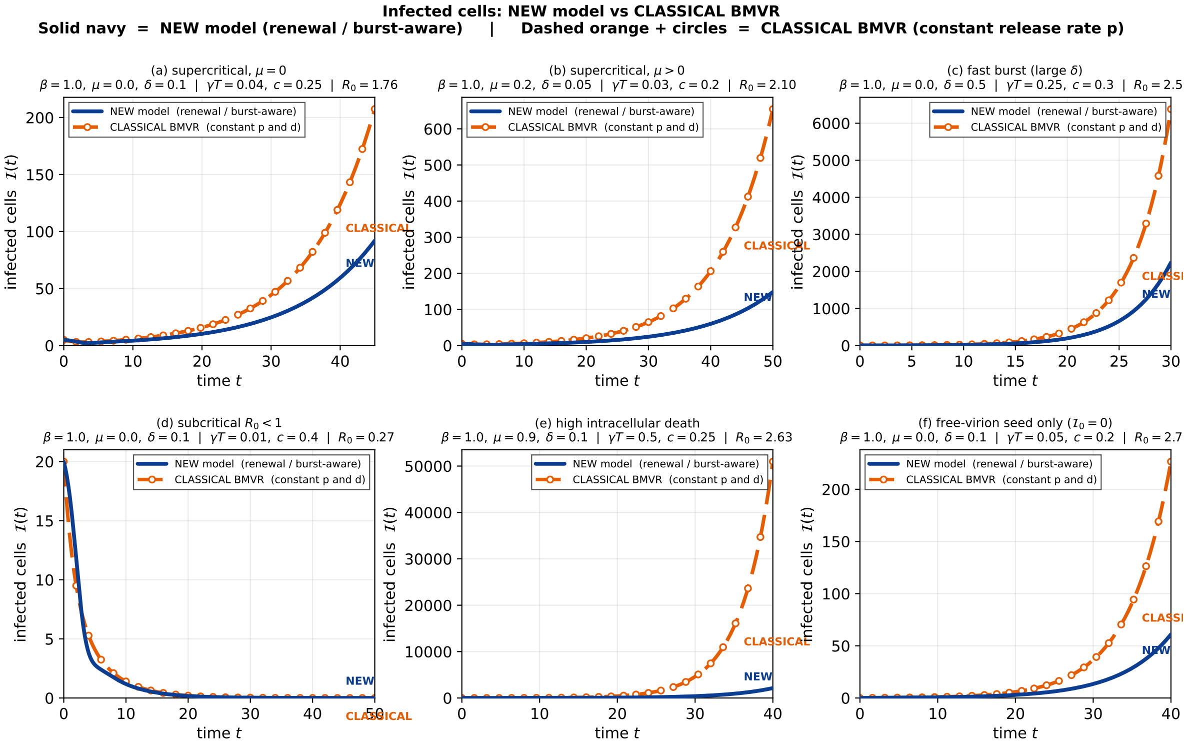

Figure 3: Infected-cell trajectories $\mathcal{I}(t)$ for the same six scenarios. Same line code as Figure 2: **solid navy = NEW**, **dashed orange = CLASSICAL**.

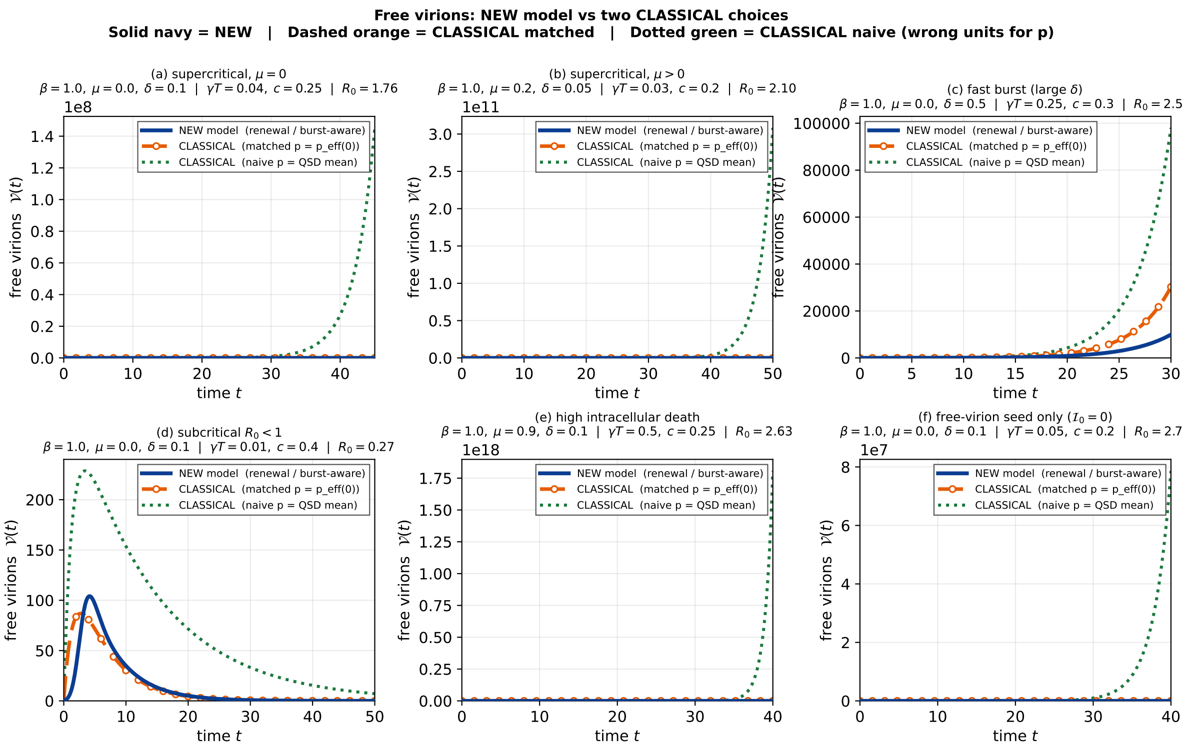

Figure 4: Optional third curve (dotted green): CLASSICAL BMVR with the dimensionally inconsistent Chapter 4 choice $p = \langle X \rangle_{\mathrm{QS}}$. Still read as: **solid navy = NEW**; **dashed orange = CLASSICAL matched**; dotted green = CLASSICAL naive.

6.3 Where matching fails

Even with p and $d_{\mathcal{I}}$ chosen to match mean lifetime output, the CLASSICAL ODEs cannot reproduce the NEW-model transients: early on, young cells dominate and release is closer to the young-cell limit $p \to \delta$; later, if the epidemic grows, generation-time structure (and hence the operative $p_{\text{eff}}(r)$) departs from the $r = 0$ value. The relative discrepancy

$$\frac{|\mathcal{V}^{\text{NEW}}(t) - \mathcal{V}^{\text{CLASSICAL}}(t)|}{\max_s \mathcal{V}^{\text{NEW}}(s)} \tag{26}$$

is plotted in Figure 5. In supercritical regimes the error is often order-one or larger by the end of the plotted window; in the high-death panel it is especially severe because V_∞ is small while the age-structure of release remains strongly non-exponential.

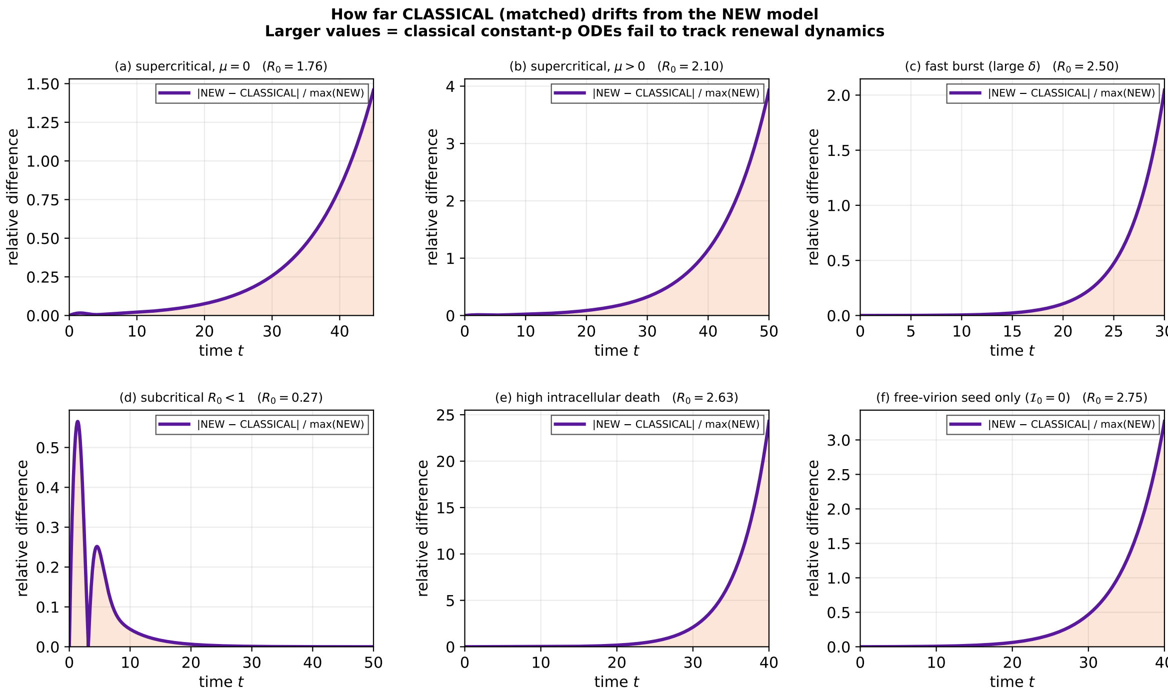

Figure 5: Relative free-virion discrepancy $|\text{NEW} - \text{CLASSICAL}|/\max(\text{NEW})$. Large values mean the classical constant-p ODEs fail to track the new model.

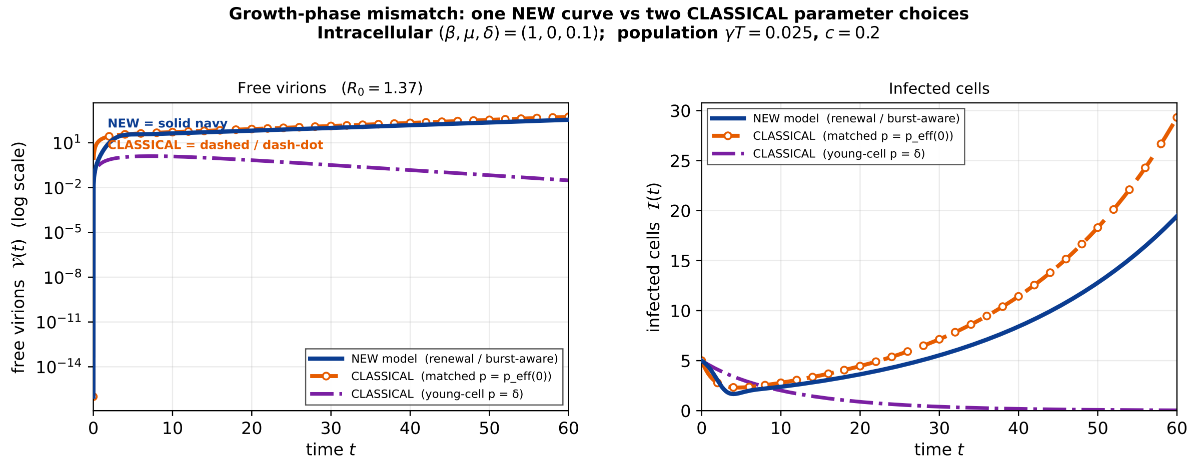

Figure 6: Growth-phase mismatch for one supercritical set ($\beta = 1$, $\mu = 0$, $\delta = 0.1$). **Solid navy = NEW model**; **dashed orange = CLASSICAL** with r=0 matching; **dash-dot purple = CLASSICAL** with young-cell limit $p = \delta$, $d_{\mathcal{I}} = \mu + \delta$. Neither classical choice tracks the new curve over the whole trajectory. Left panel logarithmic in $\mathcal{V}$.

Reading the plots.

- Always: **solid navy = NEW**, **dashed orange = CLASSICAL**.
- **CLASSICAL matched** ($p = p_{\text{eff}}(0)$) is the fairest constant-parameter comparison. It often gets the order of magnitude, but not the shape (early lag, peak timing, late growth rate).
- **CLASSICAL young-cell** ($p = \delta$) is better only briefly, then under-produces.
- **CLASSICAL naive** ($p = \langle X \rangle_{\text{QS}}$) treats a count as a rate and is systematically wrong in scale (Figure 4).
- Discrepancies grow with supercritical R_0 and with strong age-structure in the kernels (slow bursting, nonzero μ).

These numerical facts are exactly what the analytic reduction of Section 7 predicts: constant $(p, d_{\mathcal{I}})$ can match one growth rate r exactly, but not an entire transient that samples a continuum of ages.

7 Effective parameters, R_0, and growth-phase dependence

7.1 Mean productive lifetime

Result 7.1.

$$\mathbb{E}[T_{\text{prod}}] = \int_0^\infty \widehat{I}(t)\,\mathrm{d}t = \frac{1}{\beta}\log\Big(\frac{a}{a-1}\Big) = \frac{1}{\beta}\log\langle X \rangle_{\text{QS}}. \tag{27}$$

When $\mu = 0$ this is $\beta^{-1}\log(1 + \beta/\delta)$. The productive lifetime is the time exponential growth at rate β takes to reach the quasi-stationary level.

7.2 Exact reduction in the exponential phase

Substitute $i(t) = i_0 e^{rt}$ into the renewal system and compare with the classical BMVR. Writing $\widetilde{f}(r) = \int_0^\infty e^{-r\alpha} f(\alpha)\, d\alpha$:

Result 7.2 (Effective BMVR parameters)**.**

$$p_{\text{eff}}(r) = \frac{\delta\, \widetilde{K}(r)}{\widetilde{\widehat{I}}(r)}, \qquad d_{\mathcal{I},\text{eff}}(r) = \frac{1}{\widetilde{\widehat{I}}(r)} - r. \tag{28}$$

This reduction is *exact* for exponential solutions (not for transients): substituting into the BMVR eigenvalue relation recovers the renewal characteristic equation. Limits:

$$r = 0: \quad p_{\text{eff}} = \frac{\beta\, V_\infty}{\log(a/(a-1))}, \quad d_{\mathcal{I},\text{eff}} = \frac{\beta}{\log(a/(a-1))}; \tag{29}$$

$$r \to \infty: \quad p_{\text{eff}} \to \delta, \qquad d_{\mathcal{I},\text{eff}} \to \mu + \delta. \tag{30}$$

The young-cell limit is intuitive: a newly infected cell holds one particle and bursts at rate δ.

7.3 R_0 is unchanged by bursting

Result 7.3. *The characteristic equation is*

$$r + c = \gamma T\, \delta \widetilde{K}(r), \qquad \textit{hence} \qquad R_0 = \frac{\gamma T\, V_\infty}{c}.$$

At fixed total per-cell output V_∞, replacing constant budding by bursting leaves R_0 exactly unchanged. Bursting alters generation times — hence the growth rate r and the extinction probability — but not the threshold.

7.4 p is not a constant of the pathogen

Because p_{eff} decreases from $p_{\text{eff}}(0)$ toward δ as r grows, the effective release rate is smaller during acute exponential growth (young cells, low loads) than at set-point. At $(\beta, \mu, \delta) = (1, 0, 0.1)$ one has $p_{\text{eff}}(0) \approx 4.59$ against $p_{\text{eff}}(\infty) = 0.1$.

Result 7.4 (Prediction)**.** *BMVR fits to acute-phase data and to set-point data should yield systematically different p, in a direction and by an amount the theory specifies. Equivalently: the eclipse-like delay usually inserted by hand into within-host models emerges here from the intracellular dynamics, with no extra parameter.*

7.5 Identifiability (a useful negative result)

Three intracellular rates (β, μ, δ) map onto two BMVR parameters $(p, d_{\mathcal{I}})$ at any given r. The map is not invertible. Recovering the intracellular rates therefore needs a third observable, for example:

1. burst-size dispersion (the geometric ratio $1/a$);
2. fraction of infected cells that clear without bursting (b);
3. shape of the release-time density $\varphi = \delta J$.

Any one of these plus $(p, d_{\mathcal{I}})$ over-determines (β, μ, δ) and gives a consistency test.

8 The flooding advantage

This is the strongest single biological result: it answers, exactly, when bursting helps a pathogen establish at low numbers.

8.1 Offspring of one infected cell

Each released particle independently infects with probability $q = \gamma T/(\gamma T + c)$. The offspring generating function of one bursting cell is

$$G_{\text{off}}(z) = b + \frac{\delta}{\beta}\frac{y}{a-y}, \qquad y = 1 - q + qz, \tag{31}$$

with mean $m = qV_\infty$ ($= R_0$ when $\gamma T \ll c$).

8.2 Extinction probability in closed form

$G_{\text{off}}(z) = z$ is quadratic in y. The non-trivial root is

$$z_{\text{ext}}^{\text{burst}} = \frac{a(1-q+qb) - 1 + q}{q}. \tag{32}$$

8.3 Comparison with matched-R_0 budding

Comparator: a BMVR cell with exponential lifetime producing at constant rate, matched so that total mean output equals V_∞, with the same q. Its offspring number is geometric with mean R_0, so $z_{\text{ext}}^{\text{bud}} = 1/R_0$.

Write $L := a(1-b) = a - \mu/\beta$, so $V_\infty = L/(a-1)$. Then:

Result 8.1 (Flooding advantage — main theorem)**.** *For $R_0 > 1$,*

$$z_{\text{ext}}^{\text{burst}} - z_{\text{ext}}^{\text{bud}} = (L-1)\Big(\frac{1}{R_0} - 1\Big). \tag{33}$$

Hence bursting strictly lowers extinction probability relative to matched-R_0 budding if and only if $L > 1$, i.e.

$$\boxed{\frac{1}{\delta} < \frac{1}{\beta} + \frac{1}{\mu}}. \tag{34}$$

When $L = 1$ the two extinction probabilities are identical for every q. *When $\mu = 0$ the condition holds for all $\delta > 0$: bursting always wins.*

Interpretation. Bursting concentrates output into one batch, which lowers offspring variance relative to the geometric output of a budding cell. But it also carries the all-or-nothing hazard that the intracellular population dies out first (probability b) and the cell releases nothing. The condition says the advantage survives exactly when the burst timescale $1/\delta$ is shorter than the harmonic-type combination of birth and death timescales.

Numerical checks (branching simulation).

(β, μ, δ)	q	z^{burst}	z^{bud}	Winner
$(1, 0, 0.1)$	0.2	0.400	0.455	bursting ($L > 1$)
$(1, 0.9, 0.1)$	0.9	0.935	0.844	budding ($L < 1$)
$(1, 0.5, 1/3)$	0.6	0.833	0.833	tie ($L = 1$)

Remark 8.2. The comparison holds the offspring *mean* fixed and is therefore a statement about branching structure only. It says nothing about nonlinear effects (immune saturation, local depletion) of the kind Komarova's "all run out at once" intuition concerns. Both mechanisms should be named and separated in any write-up.

9 Further consequences

9.1 Multiplicity of infection

For a cell founded by k particles,

$$\widehat{I}_k(a) = I(a)^k - D(a)^k, \qquad g_k(a) = \delta\big[k\,K(a) + k(k-1)J(a)^2\big].$$

The release kernel is *superlinear* in k, so MOI matters even in the mean — a prediction the classical BMVR cannot express, and one that single-cell or low-MOI assays can test.

9.2 Two-type process and the eclipse phase

Chapter 5 already gives the exact finite-time no-catastrophe probability $S(t)$ for a two-type BDC with type-specific rates (δ_1, δ_2). The regime $\delta_1 = 0 < \delta_2$ (GATE: no burst before conversion) has $S'(0) = 0$ — a zero initial hazard followed by a rising one. That is precisely the eclipse-phase signature that within-host models normally insert by hand; here it emerges mechanistically, with eclipse duration set by the conversion rate. Completing the two-type release and survival kernels is the natural next technical step and would make Chapters 4 and 5 one continuous story.

9.3 Single-cell assay predictions (lytic BDC)

A limiting-dilution assay of the Hataye type measures single-lineage release. For a pure one-type BDC, predictions include:

1. a fraction b of wells show no release ever;
2. cumulative release per well is a step: zero, then one terminal jump;
3. jump times follow density $\delta J(t)/(1 - b)$;
4. jump sizes are geometric with ratio $1/a$ (same law as the QSD);
5. between-well variance equals $\mathrm{Var}(W_\infty)$, which the deterministic BMVR cannot produce.

Those predictions are appropriate for genuinely lytic systems (*Y. pestis*, *F. tularensis*, lytic phage). For HIV they are too strong: see the next section.

10 Distinction with HIV: what Hataye et al. show, and a candidate model

The core BDC and renewal results above are written for pathogens whose intracellular life ends in a **load-proportional catastrophe** (lyse and dump). HIV is different. It **buds** from the plasma membrane; the infected cell is not required to rupture and empty itself in one event. Single-cell and limiting-dilution data nevertheless show something that *looks* surge-like: long quiet periods, then a concentrated bout of release. The right modelling response is not to force HIV into pure BDC, nor to keep classical constant-p BMVR, but to use an **eclipse-structured, short productive pulse** with the same renewal free-virus coupling.

10.1 What the Hataye et al. (2019) paper shows

Hataye, Casazza, Best, Perelson, Koup and co-authors (*Cell Host & Microbe* 26:748–763, 2019) couple limiting-dilution latency-reactivation cultures to stochastic models. The findings that matter for equation choice are the following.

On initial release (viral inhibition cultures; no de novo infection).

1. Timing and magnitude of first HIV RNA release from reactivated latently infected CD4 T cells are **highly variable** (total yield, delay to first detection, detection duration).
2. Mean release is delayed: first detection most often around day 3, with a **sudden bulk of production around days 3–5** — not a flat continuous drip from day 0.
3. A simple productive-cell model (release at constant rate p, death at rate δ, total yield geometric with mean p/δ) is **under-dispersed** relative to the data (Fano factor too small).
4. A single exponential eclipse $E \to I$ fits the delay poorly (mode at day 1). A **multi-stage eclipse** (Erlang-like chain of n compartments; they favour $n \approx 5$) fixes the mean timing.
5. Multi-stage eclipse alone still under-disperses. Transcriptional ON/OFF toggling does not fix dispersion under realistic CD4 division/death rates ($\rho \approx \mu \approx 0.5$/day from CFSE and population data).
6. Their best structure (**Model 3**) uses **two distinct eclipse paths** (high- and low-release potential), multi-stage delay, and division/death in eclipse. Then:
   - many stimulated lineages release **nothing** detectable ($f_{\mathrm{HIV}} \approx 0.41$ of single LICs give detectable virus in their fit);
   - most productive sojourns in I last on the order of **about a day**;
   - multi-day "sustained" detection is largely from **proliferation in eclipse** then staggered short $E \to I$ transitions of descendants — not one cell budding steadily for a week.

On establishment (outgrowth cultures; de novo infection allowed).

1. Release of replication-competent virus often fails to establish exponential spread.
2. Establishment probability as a function of the expected number of virus-releasing LICs per well is **sigmoid**, not the concave curve of independent per-cell success. They interpret this as an **Allee effect** (synergy at low census).

3. A critical threshold sits near $\Lambda \approx 2.3$ LICs, corresponding on average to roughly 5100 **detected HIV RNA copies** of initial release (with their establishment definition $\geq 2 \times 10^5$ RNA copies).
4. For exactly one stimulated LIC, their fitted dynamics predict establishment only on the order of $\sim 2\%$ of the time — consistent with the known gap between intact provirus and quantitative viral outgrowth.

Reading for this document. Hataye et al. therefore support three modelling decisions and reject two others:

Supported for HIV	Not supported as literal HIV
Multi-stage eclipse (delay)	Pure one-type BDC terminal catastrophe
Short productive pulse then cell death	Memoryless constant-p BMVR from $t = 0$
High stochasticity; many silent lineages	Independent branching alone near threshold
Renewal / stage structure for free virus	QSD = geometric burst size as an HIV law
Allee / synergistic establishment	

The visual "surge" is real, but in their best model it is **delay + brief constant-p production + latent lineage dynamics**, not a single lytic dump of intracellular load.

10.2 Biological distinction from the lytic BDC

	Lytic BDC (this document, core)	HIV (Hataye-guided)
Release mechanism	Catastrophe at rate δX	Budding while productive
End of production	Terminal; process absorbed	Short I lifetime; cell dies
Intracellular load X	Explicit birth–death state	Not required as primary state
Quiet period	Until first catastrophe	Multi-stage eclipse (+ latency)
Sustained multi-day RNA	One large dump (or nothing)	Often staggered short pulses
Establishment theory	Independent offspring OK as baseline	Allee synergy near threshold
Natural hosts	*Y. pestis*, *F. tularensis*, phage	CD4 T cell latency / rebound

10.3 Candidate equations for the HIV case

The following is a compact, Hataye-aligned system suitable for coupling to the renewal free-virus layer of Result 5.1. Notation is chosen to avoid collisions with the BDC rates (β, μ, δ).

Latent / eclipse / productive stages (one LIC lineage)

Let L be a stimulated latently infected cell (or the start of reactivation). Two eclipse pathways A (high release potential) and B (low, brief) run in parallel; pathway A is multi-stage.

$$\boxed{\begin{aligned} &L \xrightarrow{\alpha_A} E_1^{(A)}, && L \xrightarrow{\alpha_B} E^{(B)}, && (35)\\ &E_j^{(A)} \xrightarrow{a} E_{j+1}^{(A)}, \quad j = 1, \dots, n-1, && E_n^{(A)} \xrightarrow{a} I^{(A)}, && (36)\\ &E^{(B)} \xrightarrow{a_B} I^{(B)}. && && (37)\end{aligned}}$$

In each eclipse compartment, cells may divide or die without releasing virus:

$$E \xrightarrow{\rho} E + E, \qquad E \xrightarrow{\mu_E} \varnothing \qquad \text{(same rates on } E_j^{(A)} \text{ and, if desired, on } E^{(B)}\text{)}. \tag{38}$$

Hataye et al. take $\rho \approx \mu_E \approx 0.5\,/\text{day}$ from CFSE and bulk population stability under stimulation. Productive cells release free virus (or supernatant RNA) at a constant rate and die:

$$\boxed{\begin{aligned} &I^{(A)} \xrightarrow{p_A} I^{(A)} + \mathcal{V}, && I^{(A)} \xrightarrow{\delta_I} \varnothing, && (39)\\ &I^{(B)} \xrightarrow{p_B} I^{(B)} + \mathcal{V}, && I^{(B)} \xrightarrow{\delta_I} \varnothing, && (40)\end{aligned}}$$

with $p_A \gg p_B$ (or p_B small and $I^{(B)}$ short-lived) to match the bimodal release distribution. Free virus / RNA in culture or plasma decays

$$\mathcal{V} \xrightarrow{c} \varnothing. \tag{41}$$

(In their inhibition cultures c is set by measured supernatant RNA half-life; in vivo c is the usual plasma clearance.)

Mean-field stage ODEs (large Λ). Writing E_j for pathway A and suppressing the dual path for display,

$$\frac{\mathrm{d}E_1}{\mathrm{d}t} = \alpha_A L_0\,\delta_0(t) + \rho E_1 - (\mu_E + \rho + a)E_1, \tag{42}$$

$$\frac{\mathrm{d}E_j}{\mathrm{d}t} = aE_{j-1} + \rho E_j - (\mu_E + \rho + a)E_j, \quad j = 2, \dots, n, \tag{43}$$

$$\frac{\mathrm{d}I}{\mathrm{d}t} = aE_n - \delta_I I, \tag{44}$$

$$\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = p_A I - c\,\mathcal{V} \qquad \text{(plus the } B\text{-path contribution } p_B I^{(B)}\text{)}. \tag{45}$$

(The Dirac source stands for an initial cohort; for ongoing infection replace it by an incidence term as below.) Stochastic simulation of the same transitions (Gillespie) is what Hataye et al. use to match single-well dispersion.

Coupling to free virus and new infections

When de novo infection is allowed, incidence should feed new eclipse cells, not instant productively infected cells:

$$\boxed{\begin{aligned} &i(t) = f(\mathcal{V}(t), T), && && (46)\\ &\text{source into } E_1^{(A)} = \pi_A\, i(t), && \text{source into } E^{(B)} = \pi_B\, i(t), && (47)\end{aligned}}$$

with $\pi_A + \pi_B = 1$ the pathway split.

Linear infection (classical).

$$f(\mathcal{V},T) = \gamma T\,\mathcal{V}.$$

This is the right large-population limit and recovers a stage-structured BMVR.

Allee / synergistic infection (Hataye establishment). Near the critical threshold their data favour a sigmoid establishment law, not independent per-virion success. A minimal phenomenological upgrade is

$$f(\mathcal{V},T) = \gamma T\,\mathcal{V} \cdot \frac{\mathcal{V}}{K_A + \mathcal{V}}, \tag{48}$$

so that infection is inefficient at very low $\mathcal{V}$ and saturates to the usual mass-action form when $\mathcal{V} \gg K_A$. The scale K_A is chosen so that cumulative initial release near $\sim 5 \times 10^3$ RNA copies sits at the inflection of establishment probability (Hataye's empirical threshold). More mechanistic forms (cooperative entry, restriction-factor saturation, local multiplicity) can replace (48); the point is that **linearity in $\mathcal{V}$ alone is not faithful near establishment**.

Renewal kernels induced by the HIV stage model

Exactly as in the lytic case, the population free-virus equation is a convolution against an age kernel — only the kernel changes. Let $g_{\mathrm{HIV}}(a)$ be the expected release rate at age a after a single infection (or reactivation), and $S_{\mathrm{HIV}}(a)$ the probability of still being in a productive (or not-yet-cleared) lineage. Then the same structural system as Result 5.1 holds:

$$\boxed{\begin{aligned} \mathcal{I}(t) &= \mathcal{I}_0 S_{\mathrm{HIV}}(t) + (i * S_{\mathrm{HIV}})(t), && (49)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \mathcal{I}_0\, g_{\mathrm{HIV}}(t) + (i * g_{\mathrm{HIV}})(t) - c\,\mathcal{V}(t). && (50) \end{aligned}}$$

For the stage model above, g_{HIV} is *not* δK from the BDC. It is the density of productive sojourn times times p_A (and p_B), with the Erlang eclipse supplying the delay. In particular $g_{\mathrm{HIV}}(a) \approx 0$ for small a, then rises and falls — the mathematical expression of "nothing for a long time, then a lot."

The effective parameters of Section 7 carry over with the substitution $(\widehat{I}, g) \mapsto (S_{\mathrm{HIV}}, g_{\mathrm{HIV}})$:

$$p_{\mathrm{eff}}(r) = \frac{\widetilde{g_{\mathrm{HIV}}}(r)}{\widetilde{S_{\mathrm{HIV}}}(r)}, \qquad R_0 = \frac{\gamma T}{c} \int_0^\infty g_{\mathrm{HIV}}(a)\,\mathrm{d}a \quad \text{(linear infection).} \tag{51}$$

Under (48), R_0 remains a large-population threshold; early establishment is controlled by the Allee scale K_A as well.

10.4 What to keep from the BDC theory, and what not to claim for HIV

- **Keep:** renewal free-virus coupling; age/stage kernels; the idea that constant p is only an exponential-phase projection; stochastic early failure; eclipse as a real dynamical object (GATE / multi-stage).
- **Do not claim for HIV:** terminal catastrophe at rate δX; QSD = geometric burst size; pure one-shot cumulative release as the only single-cell trajectory; independent branching as the full establishment story near the Hataye threshold.
- **Flooding comparison:** still meaningful as a *linear* baseline (matched mean yield, compare variance of offspring). Hataye adds that near $\sim 5 \times 10^3$ RNA copies, **synergy can dominate** variance effects. Any HIV write-up should separate the two mechanisms.

10.5 Suggested role in a paper or thesis chapter

1. Core publishable BDC + renewal BMVR results (lytic end of the spectrum).
2. One section or companion note: **HIV is not BDC**; Hataye Model 3 as the empirical intermediate; equations (35)–(50).
3. Numerically extract $g_{\mathrm{HIV}}(a)$ from the stage model (or from Hataye's daily release curves) and overlay against constant-p BMVR exactly as in Section 6.
4. Establishment: independent branching vs Allee form (48), with the critical release scale taken from Hataye et al.

That keeps the mathematical core honest for lytic pathogens while giving HIV a faithful, equation-level treatment grounded in the paper's data and models.

11 Reset-catastrophe process and its renewal BMVR

This section is self-contained. It records an intermediate intracellular model between pure continuous budding and absorbing BDC, and shows that it induces **the same renewal BMVR structure** as Section 6 / Result 5.1, with different kernels only.

Biological status (one sentence). Interpret X as a local assembled pool (not whole-cell contents): full dump is a mathematical extreme of batch release; HIV budding is better thought of as many small exports, with this model as a tractable multi-pulse caricature on the budding–bursting spectrum.

11.1 Intracellular process (one productively infected cell)

Age $a = 0$ at the start of productive infection. Load $X_a \in \{0, 1, 2, \dots\}$. While the cell is alive:

Event	Jump	Rate
Birth	$n \to n+1$	βn
Death / degradation	$n \to n-1$	μn
Immigration (optional)	$n \to n+1$	ν
Dump (reset catastrophe)	$n \to 0$, release $+n$ extracellular	δn

The cell itself dies at a separate rate $d_{\mathcal{I}}$ (cytopathic / immune), independent of X in the simplest case. On cell death the process stops. Unlike absorbing BDC, a dump does **not** end the lineage: the pool resets to 0 and may re-accumulate, so one cell can release **many** times before death.

11.2 Kernels

Dump intensity in state n is δn and the dump size is n, so the expected release rate at age a is $\delta \mathbb{E}[X_a^2 \mathbf{1}_{\{\text{alive}\}}]$. With independent cell death at rate $d_{\mathcal{I}}$,

$$\boxed{S(a) = e^{-d_{\mathcal{I}} a}, \qquad g(a) = \delta\, e^{-d_{\mathcal{I}} a}\, \mathbb{E}[X_a^2],} \tag{52}$$

where $\mathbb{E}[X_a^2]$ is computed for the reset process (no absorption at a rupture state). The structural formula $g \propto \mathbb{E}[X^2]$ matches BDC; the second moment differs because X restarts after each dump instead of ending in R.

Linear rates keep moments tractable: $\mathbb{E}[X_a]$ and $\mathbb{E}[X_a^2]$ obey a closed ODE system under the process generator (birth, death, immigration, and the reset jump). Then g is explicit once those ODEs are solved.

11.3 Renewal BMVR (same skeleton as BDC)

With target cells T fixed, infection rate γ, clearance c, and incidence $i(t) = \gamma T\,\mathcal{V}(t)$:

$$\boxed{\begin{aligned} i(t) &= \gamma T\,\mathcal{V}(t), && (53)\\ \mathcal{I}(t) &= \mathcal{I}_0\,S(t) + \int_0^t i(t-\alpha)\,S(\alpha)\,\mathrm{d}\alpha, && (54)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \mathcal{I}_0\,g(t) + \int_0^t i(t-\alpha)\,g(\alpha)\,\mathrm{d}\alpha - c\,\mathcal{V}(t), && (55)\end{aligned}}$$

with kernels (52). In convolution form: $\mathcal{I} = \mathcal{I}_0 S + i * S$ and $\mathcal{V}' = \mathcal{I}_0 g + i * g - c\mathcal{V}$.

This is identical in form to Result 5.1. Only (S, g) change.

11.4 Effective parameters and mature-cell limit

For exponential incidence $i \propto e^{rt}$,

$$p_{\text{eff}}(r) = \frac{\widetilde{g}(r)}{\widetilde{S}(r)}, \qquad d_{\mathcal{I},\text{eff}}(r) = \frac{1}{\widetilde{S}(r)} - r. \tag{56}$$

With $S(a) = e^{-d_{\mathcal{I}} a}$ one has $\widetilde{S}(r) = 1/(r + d_{\mathcal{I}})$, hence $d_{\mathcal{I},\text{eff}} \equiv d_{\mathcal{I}}$ and $p_{\text{eff}}(r) = (r + d_{\mathcal{I}})\widetilde{g}(r)$. The linear-infection threshold is

$$R_0 = \frac{\gamma T}{c}\int_0^\infty g(a)\,\mathrm{d}a = \frac{\gamma T \cdot \mathbb{E}[\text{lifetime output per cell}]}{c}. \tag{57}$$

Lifetime yield is finite only if $d_{\mathcal{I}} > 0$ (otherwise recurrent dumps can continue forever).

Mature-cell limit. If X admits a stationary law π on $\{0, 1, \ldots\}$ and $d_{\mathcal{I}}$ is small,

$$p_\infty = \lim_{a\to\infty} \frac{g(a)}{S(a)} = \delta\,\mathbb{E}_\pi[X^2].$$

A long-lived infected cell then behaves as classical constant-p BMVR with that p. Transients (first dumps, young cells) still require the full kernel $g(a)$.

11.5 Comparison with absorbing BDC

	Absorbing BDC	Reset catastrophe
After dump	process ends in R	$X \to 0$, continues
Dumps per cell	≤ 1	many (until cell death)
Survival kernel	$\widehat{I}$ (no burst / not extinct)	typically $e^{-d_{\mathcal{I}}a}$
Release kernel	$\delta K_{\mathrm{BDC}}(a)$	$\delta e^{-d_{\mathcal{I}}a}\mathbb{E}[X_a^2]$
Population equations	renewal BMVR	**same form**
Mature limit	not constant budding	$p \to \delta \mathbb{E}_\pi[X^2]$

Optional eclipse. Set $g(a) = 0$ for $a < \tau_e$ (or Erlang-delay the start of the reset process); keep $S(a) = e^{-d_{\mathcal{I}}a}$. Equations (53)–(55) are unchanged.

Takeaway. Any intracellular model that supplies (S, g) induces renewal BMVR. The reset-catastrophe process does so with a multi-pulse release kernel and a classical budding limit for mature cells, while remaining in the same linear-rate / moment-closed analytic class as BDC.

12 HIV intracellular mean growth: linear vs exponential

This section records a modelling decision for coarse-grained HIV intracellular dynamics: whether mean load grows *exponentially* (birth–death) or *linearly* (template-limited production), and what that implies for a minimal regime. Export / extracellular release is left open.

12.1 The dichotomy

Let $m(a) = \mathbb{E}[X_a]$ be mean intracellular load (Gag, genomes, or assembly-ready units) after productive infection begins.

Regime	Mean law (schematic)	Mechanism
Mean-exponential	$m' = r\,m$	each unit generates more units (branching / BDC)
Mean-linear	$m' = \nu - \mu m$ (or $m' \approx \nu$)	production set by fixed templates / machinery

Absorbing BDC and pure birth–death are exponential-mean. Immigration / constant production (and the productive phase of classical BMVR) are linear-mean.

12.2 What HIV biology favours

HIV does **not** amplify infectious units inside the cell by branching:

- After integration, the DNA template count is typically **one or a few proviruses** — not an exponentially growing genome pool.

- Late genomic RNA is made by host Pol II from that fixed DNA, not by cytoplasmic RdRp-style genome cascading.
- Gag and other structural products are translated from mRNA; they do not self-replicate. Production is immigration-like: rate set by mRNA and ribosomes, minus degradation / packaging / export.

So pure $m' = rm$ birth–death is a **poor** literal description of HIV intracellular load. It fits phage genome races, intracellular bacterial division, or some cytoplasmic RNA viruses better.

The main nonlinearity is **Tat positive feedback**: transcription rate rises from a low basal level toward a saturated high state. During the switch, product counts can rise faster than a pure linear ramp (even looking briefly super-linear) because the *production rate* is increasing — not because each unit is a branching particle. Once fully ON, production is well approximated as roughly **constant** (or fluctuating about a high mean), hence mean accumulation is linear / saturating.

Confidence.

- **High:** not pure exponential-mean birth–death of virion-like units.
- **Medium–high:** main productive window is closer to mean-linear (template-limited) than exponential.
- **Medium:** exact shape of $m(a)$ (linear vs saturating vs sigmoidal) depends on species counted and on the Tat phase; few studies cleanly fit e^{rt} vs t to single-cell intracellular load.

12.3 Does the Tat intermediate exceed the late linear rate?

Tat-controlled transcription is saturating in Tat:

$$\text{rate} = r_0 + r_{\max} f(\text{Tat}), \qquad f \text{ increasing, } f \to 1.$$

The late fully-ON rate is the **ceiling**. The intermediate phase **climbs toward** that ceiling; in the mean, it does **not** systematically exceed it.

Caveats (different observables):

- Stochastic transcriptional bursts can spike above the *mean* late rate (noise, not a higher intermediate phase).
- **Net** accumulation $\mathrm{d}X/\mathrm{d}t$ can peak mid-course if export turns on late, even when production is still rising or flat — that is export balance, not Tat overshoot.

For coarse modelling, treat Tat as acceleration **up to** the late rate, not a regime hotter than late ON.

12.4 Recommended coarse-grained regime

A fair, defensible minimal regime for HIV intracellular dynamics is:

$$\boxed{\textbf{eclipse} \longrightarrow \textbf{linear (immigration-type) accumulation of } X.} \tag{58}$$

1. **Eclipse:** negligible late product / no infectious export. Prefer a fixed or multi-step (Erlang) delay so first activity is not peaked at $t = 0^+$.
2. **Productive phase:** $m' \approx \nu - \mu m$ (or $m' \approx \nu$ while accumulating), with ν the late fully-ON production rate. Tat switch-on may be folded into a slightly longer eclipse plus an abrupt start of ν.
3. **Export:** undecided at this stage (constant rate, linear in X, batch, ...). The linear-vs-exponential choice for *intracellular* accumulation does not force the export law.

This matches fixed provirus templates, Hataye-type delay then productive window, and the mean-linear assessment above. It is **not** absorbing BDC, and it does not claim every molecular species is literally linear — only that one coarse stock after eclipse is a fair reduction.

Model ranking (HIV intracellular mean growth).

Model	Fit as HIV intracellular growth
Birth–death / absorbing BDC ($m' = rm$)	poor as literal mechanism
Immigration + export/dump ($m' \approx \nu - \cdots$)	much better caricature
Eclipse then constant production rate ν	best simple default
Tat ON/OFF + immigration	best simple mechanistic default

Takeaway. Coarse-grained HIV: **eclipse, then mean-linear intracellular production**; export to be chosen later. Do not default to exponential-mean BDC load growth inside the cell. Tat adds a switch up to the late rate, not a systematic overshoot of it.

13 Building BMVR piece by piece: five cases

This section lays out a modular route from the classic Basic Model of Viral Replication (BMVR) to models with eclipse and with an explicit intracellular stock. Target-cell count T is held constant (simplified BMVR). Free-virion clearance is c; infection rate is γ. Notation for population counts: $\mathcal{I}$ (productively infected cells), $\mathcal{V}$ (free virions), and where needed E (eclipse cells) and Q (total intracellular load summed over infected cells).

Modular programme.

1. **Case 1 — Classic BMVR.** Linear production with *immediate* release at constant budding rate p per cell. No intracellular stock, no delay.
2. **Case 2 — Eclipse + classic release.** Same as Case 1, but new infections enter a non-producing eclipse class before becoming productive. (Standard in the literature.)
3. **Case 3 — Intracellular stock + continuous export.** Linear accumulation into a compartment Q; continuous release from that compartment into $\mathcal{V}$ (no eclipse in the main write-up; eclipse stacks as in Case 2).
4. **Case 4 — Intracellular stock + proportional full clear.** Same accumulation as Case 3, but release is by dump events that clear the pool at a load-proportional rate (reset catastrophe at the mean-field level). No eclipse.

5. **Case 5 — Eclipse + stock + proportional full clear.** Case 4 with an eclipse class: incidence feeds E, only productively infected cells accumulate Q and dump.

Cases 3–5 are the "accumulation + structured release" layer: the *export law* and optional eclipse distinguish them; intracellular production is immigration-type (mean-linear), not birth–death.

13.1 Case 1 — Classic BMVR

Assumptions. An infected cell produces virions that are released *immediately* at constant rate p (budding). No eclipse, no stored intracellular pool. Infected cells are removed at rate $d_{\mathcal{I}}$.

$$\boxed{\begin{aligned} \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, \quad (59)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= p\,\mathcal{I} - c\,\mathcal{V}. \quad (60)\end{aligned}}$$

Reading the equations. Infection creates productively infected cells at rate $\gamma T\mathcal{V}$. Each such cell contributes p virions per unit time to the free pool; free virions clear at rate c. Production and export are collapsed into the single parameter p.

Basic reproduction number (linearised at $\mathcal{V}$ small).

$$R_0 = \frac{\gamma T\,p}{c\,d_{\mathcal{I}}}.$$

13.2 Case 2 — Eclipse phase + classic release

Assumptions. New infections enter an eclipse (or latent-productive delay) class E: infected but not yet producing. Conversion $E \to \mathcal{I}$ at rate a (mean eclipse length $1/a$). Optional eclipse death d_E; productive cells as in Case 1. Release is still immediate constant budding p from $\mathcal{I}$ only.

This extension is standard in within-host HIV modelling (delay before production; fits to acute viremia and treatment timing).

$$\boxed{\begin{aligned} \frac{\mathrm{d}E}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - (a + d_E)\,E, \quad (61)\\ \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= a\,E - d_{\mathcal{I}}\,\mathcal{I}, \quad (62)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= p\,\mathcal{I} - c\,\mathcal{V}. \quad (63)\end{aligned}}$$

Reading the equations. Incidence feeds E, not $\mathcal{I}$. Only after conversion do cells contribute to $\mathcal{V}$ at rate p. Setting $a \to \infty$ (instant exit from eclipse) and $d_E = 0$ recovers Case 1 with the same p.

R_0. A fraction $a/(a+d_E)$ of eclipse cells become productive; each then yields mean $p/d_{\mathcal{I}}$ virions, so

$$R_0 = \frac{\gamma T}{c} \cdot \frac{a}{a+d_E} \cdot \frac{p}{d_{\mathcal{I}}}.$$

Multi-stage eclipse. Replace one E by a chain $E_1 \to \cdots \to E_n \to \mathcal{I}$ (Erlang delay) for a more realistic distribution of eclipse times; the $\mathcal{V}$ equation is unchanged. Hataye-type multi-compartment eclipse is this idea at finer grain.

13.3 Case 3 — Intracellular accumulation + continuous export

Assumptions. Productively infected cells accumulate an intracellular stock (total pool Q across all such cells) by **linear production** at rate ν per cell (immigration-type; Section 12). Virions are released **continuously** from that stock: each intracellular unit is exported at per-capita rate ε (linear budding from the compartment). Cell death removes both the cell and its share of load (mean-field: load lost at rate $d_{\mathcal{I}}Q$). New infections start with empty stock.

$$\boxed{\begin{aligned} \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, && (64)\\ \frac{\mathrm{d}Q}{\mathrm{d}t} &= \nu\,\mathcal{I} - \varepsilon\,Q - d_{\mathcal{I}}\,Q, && (65)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \varepsilon\,Q - c\,\mathcal{V}. && (66)\end{aligned}}$$

Reading the equations.

- $\mathcal{I}$: same bookkeeping as Case 1.
- Q: gained by production $\nu\mathcal{I}$; lost by export εQ and by death of infected cells $d_{\mathcal{I}}Q$.
- $\mathcal{V}$: fed only by export εQ, not by a phenomenological $p\mathcal{I}$.

Mean load per productive cell is $m = Q/\mathcal{I}$ (when $\mathcal{I} > 0$). In a quasi-steady intracellular balance ($\frac{\mathrm{d}Q}{\mathrm{d}t} \approx 0$),

$$Q_* = \frac{\nu}{\varepsilon + d_{\mathcal{I}}}\mathcal{I}, \qquad p_{\text{eff}} := \frac{\varepsilon Q_*}{\mathcal{I}} = \frac{\varepsilon\nu}{\varepsilon + d_{\mathcal{I}}},$$

and Case 3 collapses to Case 1 with $p = p_{\text{eff}}$. Thus classic BMVR is the **fast-export / quasi-steady stock** limit of this model. When stock dynamics are not fast, $\mathcal{V}$ lags and depends on the history of Q.

Constant release rate per cell (variant). If instead each productive cell exports at a fixed rate ρ independent of load (only while Q is not limiting), one may close with $\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = \rho\mathcal{I} - c\mathcal{V}$ and drain Q by $\rho\mathcal{I}$, provided Q stays non-negative — a cruder law, closer to Case 1, with Q tracking unused stock. The linear law εQ is the natural default for "release from the compartment."

With eclipse (Case 2+Case 3). Incidence enters E; only $\mathcal{I}$ produces into Q:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \gamma T\,\mathcal{V} - (a + d_E)\,E, \tag{67}$$

$$\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} = a\,E - d_{\mathcal{I}}\,\mathcal{I}, \tag{68}$$

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \nu\,\mathcal{I} - \varepsilon\,Q - d_{\mathcal{I}}\,Q, \tag{69}$$

$$\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} = \varepsilon\,Q - c\,\mathcal{V}. \tag{70}$$

Eclipse cells do not contribute to Q or $\mathcal{V}$ until conversion.

R_0 (Case 3 without eclipse). Lifetime output per cell equals production integrated against survival, $\nu/d_{\mathcal{I}}$, if every produced unit is eventually either exported or lost at death; more carefully, the export fraction is $\varepsilon/(\varepsilon + d_{\mathcal{I}})$, so mean virions per cell $= \nu\varepsilon/(d_{\mathcal{I}}(\varepsilon + d_{\mathcal{I}})) \cdot d_{\mathcal{I}}/\varepsilon$ simplifies to

$$\frac{\nu}{d_{\mathcal{I}}} \cdot \frac{\varepsilon}{\varepsilon + d_{\mathcal{I}}}, \qquad R_0 = \frac{\gamma T\,\nu\,\varepsilon}{c\,d_{\mathcal{I}}\,(\varepsilon + d_{\mathcal{I}})},$$

matching Case 1 with $p = \varepsilon\nu/(\varepsilon + d_{\mathcal{I}})$.

13.4 Case 4 — Intracellular accumulation + proportional full clear

Assumptions. Same linear production into the stock as Case 3 (ν per productive cell). Release is **not** continuous per unit. Instead, dumps occur at a rate proportional to load: in a cell with load x, dump rate δx, and the whole pool x is transferred to $\mathcal{V}$ while the cell's stock resets to 0 (cell may survive; death is still the separate rate $d_{\mathcal{I}}$).

Mean-field closure. Let Q be total intracellular load and $\mathcal{I}$ the number of productive cells. Under a homogeneous-load closure (every cell carries $m = Q/\mathcal{I}$), the population dump rate of virions is

$$\delta\,\mathbb{E}\sum_i X_i^2 \approx \delta\,\mathcal{I}\,m^2 = \delta\,\frac{Q^2}{\mathcal{I}},$$

and the same quantity is the rate at which load leaves Q. (Exact stochastic models use $\delta\mathbb{E}[X^2]$ per cell; this is the standard mean-field analogue.)

$$\boxed{\begin{aligned} \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, &(71)\\ \frac{\mathrm{d}Q}{\mathrm{d}t} &= \nu\,\mathcal{I} - \delta\,\frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}}\,Q, &(72)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \delta\,\frac{Q^2}{\mathcal{I}} - c\,\mathcal{V}, &(73)\end{aligned}}$$

(with the convention that the nonlinear terms vanish when $\mathcal{I} = 0$).

Reading the equations.

- Production $\nu\mathcal{I}$ as in Case 3.
- The term $\delta Q^2/\mathcal{I}$ is simultaneous **export into $\mathcal{V}$** and **clearing of Q** (full dump).

- Cell death still removes load at rate $d_{\mathcal{I}}Q$; dumps do *not* remove the cell from $\mathcal{I}$ (reset, not lysis). For a lytic variant, add a sink in $\mathcal{I}$ proportional to total dump rate.

Contrast with Case 3. Continuous export is linear in Q (εQ). Full clear is **quadratic** in total load at fixed $\mathcal{I}$ ($\propto Q^2/\mathcal{I}$), i.e. superlinear in mean load m. Mature quasi-steady balance ($\frac{\mathrm{d}Q}{\mathrm{d}t} \approx 0$, $\mathcal{I}$ slowly varying) gives

$$\nu = \delta m^2 + d_{\mathcal{I}}\, m \quad \Rightarrow \quad m = \frac{-d_{\mathcal{I}} + \sqrt{d_{\mathcal{I}}^2 + 4\delta\nu}}{2\delta},$$

and effective release per cell $p_{\mathrm{eff}} = \delta m^2$, which plays the role of p in Case 1 only in that steady regime — transients and early infection need the full $(Q, \mathcal{V})$ system. For the same release law **with eclipse**, see Case 5.

R_0 (sketch). In the linearised regime near zero infection, Q is small, dumps are rare, and the first-order expansion of (72)–(73) must be taken carefully (the map $Q \mapsto Q^2/\mathcal{I}$ is singular at 0 in naive form). For invasion analysis it is safer to linearise the underlying single-cell process (expected lifetime output V_∞) and set $R_0 = \gamma T V_\infty / c$, as in the renewal framework. The ODE system (71)–(73) is aimed at bulk dynamics with $\mathcal{I}$ not near zero; use branching / renewal for establishment.

13.5 Case 5 — Eclipse + stock + proportional full clear

Assumptions. Combine Case 2 and Case 4: new infections enter a non-producing eclipse class E; only after conversion to productively infected $\mathcal{I}$ does linear production into the intracellular stock Q begin; release from Q is by load-proportional full clear (same mean-field dump as Case 4). Eclipse cells carry no stock and do not feed $\mathcal{V}$.

$$\boxed{\begin{aligned}
\frac{\mathrm{d}E}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - (a + d_E)\,E, && (74)\\
\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= a\,E - d_{\mathcal{I}}\,\mathcal{I}, && (75)\\
\frac{\mathrm{d}Q}{\mathrm{d}t} &= \nu\,\mathcal{I} - \delta\,\frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}}\,Q, && (76)\\
\frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \delta\,\frac{Q^2}{\mathcal{I}} - c\,\mathcal{V}, && (77)
\end{aligned}}$$

(with nonlinear dump terms set to 0 when $\mathcal{I} = 0$).

Reading the equations.

- (74)–(75): same eclipse bookkeeping as Case 2. Incidence creates E; conversion at rate a creates $\mathcal{I}$; optional eclipse death d_E; productive death $d_{\mathcal{I}}$.
- (76)–(77): identical stock and release law to Case 4. Only productively infected cells produce ($\nu\mathcal{I}$) and dump ($\delta Q^2/\mathcal{I}$).
- Setting $a \to \infty$ with $d_E = 0$ recovers Case 4 (instant exit from eclipse). Setting the dump aside and replacing $\delta Q^2/\mathcal{I}$ by continuous export would give eclipse+Case 3.

What eclipse changes relative to Case 4. There is an additional delay before any contribution to Q or $\mathcal{V}$. Early after a pulse of infection, E rises first, then $\mathcal{I}$ and Q, then $\mathcal{V}$ — a staged lag that Case 4 cannot produce. Quasi-steady relations for mean load $m = Q/\mathcal{I}$ among *productive* cells are unchanged from Case 4 once $\mathcal{I}$ is slowly varying; the eclipse only filters how $\mathcal{I}(t)$ is driven by $\mathcal{V}$.

R_0 (sketch). As in Case 2, a fraction $a/(a + d_E)$ of eclipse cells become productive. If V_∞ is expected lifetime free-virus output of one productive cell under the Case 4 release law, then

$$R_0 = \frac{\gamma T}{c} \cdot \frac{a}{a + d_E} \cdot V_\infty$$

in the linear-infection, single-cell sense. Bulk invasion near $\mathcal{I} = 0$ has the same caveat as Case 4 regarding the $Q^2/\mathcal{I}$ closure; prefer single-cell / renewal linearisation for thresholds.

13.6 Summary of the five cases

Case	Intracellular story	Free-virus source term
1 Classic	none (instant release)	$p\mathcal{I}$
2 Eclipse + classic	delay $E \to \mathcal{I}$, then as 1	$p\mathcal{I}$
3 Stock + continuous export	ν into Q, export εQ	$\varepsilon\, Q$
4 Stock + full clear	ν into Q, dump $\propto Q^2/\mathcal{I}$	$\delta Q^2/\mathcal{I}$
5 Eclipse + stock + full clear	$E \to \mathcal{I}$, then as 4	$\delta Q^2/\mathcal{I}$

What is fixed vs open.

- **Fixed in Cases 3–5:** production into the stock (once productive) is **linear / immigration-type**, not birth–death exponential mean growth.
- **Open / chosen by case:** release law — continuous (εQ) vs proportional full clear ($\delta Q^2/\mathcal{I}$); and whether eclipse is present (Cases 2 and 5, or stacked on Case 3).
- **Composability:** Case 5 is exactly Case 4 with Case 2's eclipse block in front.
- **Link to renewal BMVR:** each case induces kernels $S(a)$ and $g(a)$; the convolution form of Result 5.1 remains available when ODE stage structure is replaced by age structure.

Takeaway. Start from classic BMVR (Case 1), add eclipse if delay matters (Case 2), then — when release timing and stock matter — replace $p\mathcal{I}$ by an explicit compartment Q with either continuous export (Case 3) or proportional full clear (Case 4). Add eclipse in front of full clear as Case 5 when both delay and batch-like dump matter. Classic p is recovered as a quasi-steady limit of Case 3.

14 Self-excitatory release mechanisms

Cases 3–5 treat release as either continuous export or load-proportional full clear. A further biological possibility is that release is partially **self-excitatory**: one budding / dump event makes further release more likely for a short time (Hawkes-type thinking), e.g. via cooperative Gag nucleation, local membrane platforms, or ESCRT already recruited at a site.

This must be balanced against **depletion** of stock X and finite machinery: pure open-ended self-excitation is unlikely; a hybrid "excitation + refill + depletion" picture is more plausible than a classical Hawkes process alone.

Two levels of BMVR.

- **ODE BMVR:** finite system in $(\mathcal{I}, Q$ or $m, \mathcal{V})$ (and maybe E, phase variables).
- **Renewal BMVR:** $(\mathcal{I}, \mathcal{V})$ with kernels $S(a)$, $g(a)$ from a single-cell process — same skeleton as Result 5.1.

Renewal BMVR exists whenever the single-cell model defines an expected release rate $g(a)$ and survival $S(a)$. ODE BMVR needs a low-dimensional Markov state (or a moment closure).

14.1 Menu of self-excitatory models

#	Model	Excitation type	ODE BMVR?	Renewal?
1	Classical Hawkes	event history only	poor	yes (num. g)
2	Hawkes + stock X	events + depletion/refill	approx. / via 10	**yes**
3	Marked Hawkes	size-weighted	approx.	yes (num.)
4	ETAS-style	size + time kernel	approx.	yes (num.)
5	Cox + latent Z	latent excitability	yes if Z simple	yes
6	ON/OFF telegraph	sustained high phase	**yes**	yes
7	$\lambda \propto X^2$	cooperative rate, no history	**yes** (Cases 4–5)	yes
8	Nucleation biophysics	assembly cascade	hard	yes (num.)
9	Age since last release	renewal excitation	yes (stages)	yes
10	Jumping rate $r(t)$	shot-noise intensity	**yes** (mean-field)	yes
11	Spatial Hawkes	local membrane	average first	after coarse-grain
12	Full clear (Cases 4–5)	instant total cascade	**yes**	yes

How they nest. Model 2 (Hawkes + X) is the general event-history form. With $\phi \equiv 0$ and $f(x) = \varepsilon x$ it recovers continuous export (Case 3); with $\phi \equiv 0$, $f(x) = \delta x^2$ and full clear it recovers model 7 / Cases 4–5; with exponential $\phi(t) = \eta e^{-\beta t}$ it is equivalent to model 10 (Markov intensity). Model 6 is a coarse two-level approximation to high vs low release rate without continuous r.

The four models best suited to our build are detailed next: **7**, **6**, **10**, and **2**.

14.2 Model 7 — Quadratic / cooperative intensity ($\lambda \propto X^2$)

Idea. Release is a state-dependent Poisson process. Given load $X = x$,

$$\lambda(x) = \delta x^2 \quad (\text{or } \delta x(x-1)).$$

No explicit memory of past event times — only current load. "Self-excitation" means **cooperativity**: more material present makes release disproportionately more likely (nucleation / lattice / local Gag density). This is the single-cell version of the mean-field dump rate in Cases 4–5.

Single-cell transitions (productive cell).

Event	Effect	Rate
Production	$X \to X+1$	ν (linear / immigration)
Optional degradation	$X \to X-1$	μX
Dump (full clear)	$X \to 0$, release $+X$	δX^2
Cell death	stop	$d_{\mathcal{I}}$

Population BMVR. ODE form is already Cases 4–5:

$$\mathcal{V}' = \delta \frac{Q^2}{\mathcal{I}} - c\mathcal{V}, \qquad Q' = \nu \mathcal{I} - \delta \frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}} Q$$

(homogeneous closure $\mathbb{E}[X^2] \approx m^2$, $m = Q/\mathcal{I}$). Renewal form: $g(a) = \delta \mathbb{E}[X_a^2 \mathbf{1}_{\{\text{alive}\}}]$, $S(a) = e^{-d_{\mathcal{I}} a}$, then the usual convolutions.

Pros / cons. ODE and renewal both available; interpretable cooperativity without Hawkes; homogeneous $Q^2/\mathcal{I}$ ignores load variance; full clear is an extreme (unit budding at rate εX is milder).

Use when. Default bulk ODE "excitatory" model; continuum of cooperative nucleation if event history is not required.

14.3 Model 6 — Two-state telegraph / bursting release (ON–OFF)

Idea. The cell (or a membrane site) switches between OFF (little or no release) and ON (release at a high rate). Self-excitation is **phase-based**: once ON, many releases occur until OFF — a burst, not a Hawkes sum over every past event. Optional: OFF→ON rate increases with X or after a recent release.

Single-cell sketch. State (M, X) with $M \in \{\text{OFF}, \text{ON}\}$. Production at rate ν (always or only ON); OFF→ON at rate α (or $\alpha_0 + \alpha_1 X$); ON→OFF at rate β; release while ON at rate εX or constant ρ; cell death $d_{\mathcal{I}}$.

Sample path. OFF: stock may build, little free virus; ON: rapid release, stock may fall; OFF again: rebuild. Intermittent surges without requiring full clear.

Population ODE BMVR (simplest — constant ρ while ON).

$$\boxed{\begin{aligned} \frac{\mathrm{d}I_{\mathrm{OFF}}}{\mathrm{d}t} &= \gamma T\,\mathcal{V} + \beta I_{\mathrm{ON}} - (\alpha + d_{\mathcal{I}})I_{\mathrm{OFF}}, && (78)\\ \frac{\mathrm{d}I_{\mathrm{ON}}}{\mathrm{d}t} &= \alpha I_{\mathrm{OFF}} - (\beta + d_{\mathcal{I}})I_{\mathrm{ON}}, && (79)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \rho\, I_{\mathrm{ON}} - c\,\mathcal{V}. && (80) \end{aligned}}$$

(Eclipse: feed E by incidence, convert into I_{OFF}.) Quasi-steady switching yields effective budding

$$p_{\mathrm{eff}} = \rho \cdot \frac{\alpha}{\alpha + \beta},$$

recovering classic BMVR (Case 1) as a duty-cycle limit.

With stock. Track Q with production $\nu(I_{\mathrm{ON}} + I_{\mathrm{OFF}})$ (or only ON) and export εQ restricted to ON cells (requires a closure for how load is shared). Renewal BMVR: single-cell (M_a, X_a) defines g and S.

Pros / cons. Among excitatory models, one of the **best for writable ODEs**; matches intermittent release; links to Tat ON/OFF intuition; extra parameters (α, β, ρ); not true Hawkes (no decaying event kernel).

Use when. Clustered release in time with clear ODEs and a transparent limit to constant p.

14.4 Model 10 — Jumping continuous release rate $r(t)$

Idea. Keep a continuous release-rate (intensity) process $r(t) \geq 0$: jumps up on excitation events, decays toward a baseline between jumps. Export flux is $r(t)$ or $r(t)X(t)$. This is the **Markovian form of Hawkes** when the excitation kernel is exponential (shot-noise representation).

Single-cell (exponential kernel).

$$\mathrm{d}r = -\beta(r - r_0)\,\mathrm{d}t + \eta\,\mathrm{d}N_t, \tag{81}$$

$$\lambda(t) = r(t) \quad \text{or} \quad r(t)\,X_t, \qquad N \text{ has intensity } \lambda, \tag{82}$$

with X refilled at rate ν and decremented on each release. Each event adds η to r; memory fades at rate β.

Population ODE BMVR (mean-field sketch). Track $\mathcal{I}$, total load Q, total excitation $R = \sum r_i$ (mean $\bar{r} = R/\mathcal{I}$). With flux $\approx \bar{r}\,Q$ and jump input $\propto$ event rate $\approx \bar{r}\,Q$:

$$\boxed{\begin{aligned}\frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, && (83)\\ \frac{\mathrm{d}Q}{\mathrm{d}t} &= \nu\mathcal{I} - \bar{r}\,Q - d_{\mathcal{I}}Q, && (84)\\ \frac{\mathrm{d}R}{\mathrm{d}t} &= -\beta(R - r_0\mathcal{I}) + \eta\,(\bar{r}\,Q) + (\text{cell birth/death corrections}), && (85)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \bar{r}\,Q - c\,\mathcal{V}, && (86)\end{aligned}}$$

using closures $\mathbb{E}[rX] \approx \bar{r}\,m$. Renewal BMVR: $g(a) = \mathbb{E}[r_a X_a]$ (if flux $= rX$); kernels usually numerical.

Pros / cons. True self-excitation with fading memory; finite-dimensional intensity state allows ODE BMVR; bridges Hawkes and bulk models; needs closures and extra parameters (β, η, r_0); hard to identify from viral load alone.

Use when. Hawkes-type memory is desired but host-level equations should remain ODEs.

14.5 Model 2 — Hawkes process with stock X

Idea. Release times (t_i) with intensity

$$\lambda(t) = f(X_t) + \sum_{t_i<t} \phi(t - t_i), \tag{87}$$

$f(X)$ baseline from load (e.g. $f(x) = \varepsilon x$ or δx^2); $\phi \geq 0$ self-excitation kernel; on event, $X \leftarrow X - 1$ (or batch / full clear); between events, X grows at rate ν; cell dies at rate $d_{\mathcal{I}}$. This is the most literal "Hawkes for budding with biology."

Special cases.

Choice	Recovers
$\phi = 0,\ f(x) = \varepsilon x$	Case 3 continuous export
$\phi = 0,\ f = \delta x^2$, jump $X \to 0$	Model 7 / Cases 4–5
$f = \text{const},\ \phi = \eta e^{-\beta t}$, ignore X	classical Hawkes
$f = \varepsilon x,\ \phi = \eta e^{-\beta t}$	**canonical hybrid**
exponential ϕ	equivalent to Model 10

Population BMVR.

- **Renewal:** always well-defined — $g(a) = \mathbb{E}[\lambda(a) \mid \text{alive}]$, S from cell lifetime; same convolutions as BDC. Kernels typically from simulation (or from Model 10 if ϕ is exponential).
- **ODE:** not without reduction — infinite event history. Restrict to exponential ϕ and use Model 10 mean-field, or apply moment closures.

Pros / cons. Direct Hawkes language; flexible short excitation window (e.g. ESCRT); combines load, excitation, and depletion; analytics hard unless exponential ϕ; ODE BMVR only after reduction to Model 10.

Use when. Conceptual / single-cell / renewal theory with true self-excitation; for host ODEs, specialise immediately to exponential ϕ and Model 10.

14.6 Comparison of the four focal models

	7 $\lambda \propto X^2$	**6** ON/OFF	**10** jumping r	**2** Hawkes+X
Carrier	load only	discrete phase	continuous r	event history (+ load)
Memory	Markov in X	Markov in phase	Markov in r	non-Markov unless exp. ϕ
Clusters	high $X \to$ fast dumps	long ON sojourn	high-r window	aftershocks
ODE BMVR	yes (Cases 4–5)	yes (split I)	yes (mean-field R)	only via 10
Renewal BMVR	yes	yes	yes	yes
$\to$ classic p	QSS δm^2	duty cycle	QSS $\mathbb{E}[rX]$	as 10 if exp ϕ

Biological confidence (shared caveat). Cooperative assembly / nucleation (Gag lattice, scaffolds, ESCRT) makes **short clustered** release plausible (**medium–high** as a hypothesis). A classical Hawkes process with only event memory and no depletion is a poor standalone cell model (**low–medium**). Hybrid excitation + stock depletion/refill is a reasonable working hypothesis; it is **not** established as the dominant single-cell export law for HIV.

Practical ranking for this project.

1. Writable ODEs inside current Cases 3–5: Model **7**.
2. Intermittent release, simplest new ODEs: Model **6**.
3. Self-excitation with fading memory + ODEs: Model **10**.
4. Paper language "Hawkes-like budding": define Model **2**, implement ODEs as Model **10**, or renewal with numerical g.

Takeaway. Self-excitation is compatible with BMVR whenever the single cell supplies (S, g) (renewal form always). Finite ODE BMVR needs excitation carried by a few Markov states — load X (Model 7), ON/OFF (Model 6), or intensity r (Model 10) — not an unbounded event history, unless that history is reduced to exponential memory (Hawkes $\leftrightarrow$ Model 10).

15 Partial release: intermediates between continuous export and full clear

Cases 3–5 sit at two extremes of *how much* leaves the intracellular compartment when release occurs:

Extreme	Mechanism	In this document
Continuous export	units leave steadily (flux εQ)	Case 3
Full clear	event empties the whole pool	Cases 4–5

A natural intermediate is a **release event** that removes only a **random or fixed part** of the current load $X = n$: export B units with $0 \le B \le n$ (or $1 \le B \le n$), retain $n - B$. Self-excitation (Section 14) is optional and separate; here only the **release-count law** $B \mid X = n$ is at issue.

Two ingredients.

1. **When** an event fires — rate $\lambda(X)$, e.g. constant δ, linear δX, or cooperative δX^2.
2. **How many** leave — distribution of $B \mid X = n$.

15.1 Options for $B \mid X = n$

A. Boolean / independent retention (recommended default intermediate)

When an event fires at load n, each of the n units is released independently with probability q and retained with probability $1 - q$:

$$B \mid X = n \;\sim\; \text{Binomial}(n, q), \qquad X \leftarrow n - B \;\sim\; \text{Binomial}(n, 1-q). \tag{88}$$

q	Behaviour
$q \to 0$	tiny fractional clear
$q \in (0,1)$	true intermediate
$q = 1$	full clear (Cases 4–5 style)

Mean release given n: $\mathbb{E}[B \mid n] = qn$. Mechanistic reading: a release event "hits" the pool; each package succeeds independently with probability q. Generating functions stay friendly (product structure over units).

Event rate still free. Examples: $\lambda = \delta$ (per cell); $\lambda = \delta X$; $\lambda = \delta X^2$.

Mean-field BMVR (boolean, $\lambda = \delta X$). Population event structure as in Case 4, but only fraction q of load leaves:

$$\boxed{\begin{aligned} \frac{\mathrm{d}\mathcal{I}}{\mathrm{d}t} &= \gamma T\,\mathcal{V} - d_{\mathcal{I}}\,\mathcal{I}, && (89)\\ \frac{\mathrm{d}Q}{\mathrm{d}t} &= \nu\mathcal{I} - \delta q\,\frac{Q^2}{\mathcal{I}} - d_{\mathcal{I}}Q, && (90)\\ \frac{\mathrm{d}\mathcal{V}}{\mathrm{d}t} &= \delta q\,\frac{Q^2}{\mathcal{I}} - c\,\mathcal{V}. && (91) \end{aligned}}$$

Thus $q = 1$ recovers Case 4; $q \in (0,1)$ is partial clear with the same clock. If instead $\lambda = \delta$ (constant per cell), mean export is $\delta q\, Q$ and the bulk ODEs collapse to Case 3 with $\varepsilon_{\text{eff}} = \delta q$ — pulses remain visible in single-cell paths and higher moments, not in the first-moment ODE.

B. Fixed fraction

On event, release $B = \lceil fn \rceil$ or $\mathrm{round}(fn)$ with fixed $f \in (0, 1]$. Same mean as boolean with $q = f$, **less** stochasticity. Bulk mean-field matches boolean at equal mean fraction; single-cell burst-size variance is smaller.

C. Random fraction (continuous thinning)

Draw $U \sim \mathrm{Beta}(a, b)$ (or $\mathrm{Uniform}(0, 1)$); set $B = \mathrm{round}(Un)$ or thin continuously. More variable burst sizes than fixed f; boolean is the discrete "each particle flips a coin" analogue of random thinning.

D. Fixed batch size (capped)

$B = \min(k, n)$ for package size $k \geq 1$. Special cases: $k = 1$ is unit budding on an event clock (Case 3 in the frequent-event limit); k large approaches full clear when n is not huge.

E. Size-biased / prefer large

$P(B = j \mid n) \propto j$, or dump probability depending strongly on n. Less natural if units are identical; more natural if one site releases a random assembled blob.

F. Threshold then partial

Event only if $X \geq x_*$, then apply boolean q or fixed f. Combines accumulation with a trigger ("surge when enough material").

G. Short high-export window

On trigger, export at high ε for a short time, then stop. Partial clear in expectation without an explicit law for B; related to ON/OFF (Model 6) with brief ON.

15.2 Single-cell boolean process (compact)

Event	Rate	Jump
Produce	ν	$X \to X + 1$
Release event	$\lambda(X)$ (e.g. δX or δX^2)	$B \sim \mathrm{Bin}(n, q)$; $X \to n - B$; free virus $+B$
Cell death	$d_{\mathcal{I}}$	stop

Nested limits.

Choice	Recovers
$q = 1$, $\lambda = \delta X^2$	full clear, cooperative (Case 4-like)
$q = 1$, $\lambda = \delta X$	full clear, linear hazard
$0 < q < 1$, $\lambda = \delta X$	**main partial-clear intermediate**
always $B = 1$ when fire, $\lambda = \varepsilon X$	unit budding (Case 3 discrete)

15.3 BMVR feasibility

Setup	ODE BMVR	Renewal BMV
Boolean q, $\lambda = \delta$ (const)	yes ($\approx$ Case 3, $\varepsilon = \delta q$)	yes
Boolean q, $\lambda = \delta X$	yes (89)–(91)	yes
Boolean q, $\lambda = \delta X^2$	yes (Case 4 factor q, retain $1 - q$)	yes
General law $B \mid n$	yes if $\mathbb{E}[B \mid n]$ closes under a moment assumption	yes always via g

Renewal BMVR: single-cell process defines $g(a) = \mathbb{E}[\text{release rate at age } a]$ and survival $S(a)$; population equations as in Result 5.1.

15.4 Where partial clear sits relative to continuous export

Model	Single-cell path	Mean bulk ODE
Case 3 (εQ)	smooth drip	linear export
Boolean, $\lambda = \delta$, any q	rare binomial pulses	$\approx$ linear $\delta q\, Q$
Boolean, $\lambda = \delta X$, $q \in (0, 1)$	pulses of mean size qm	quadratic, strength δq
Case 4 ($q = 1$)	full empty pulses	quadratic full clear

Partial vs full is not always visible in mean ODEs if the event rate is constant (then only single-cell burst sizes and higher moments differ). It **is** visible in bulk dynamics when the event rate depends on load (e.g. $\lambda = \delta X$), and always in single-cell release-count distributions.

15.5 Recommendation

Primary intermediate: boolean (binomial) thinning with event rate $\lambda = \delta X$ (or δX^2) and release probability $q \in (0, 1]$. One transparent parameter q between full clear and mild partial clear; mechanistic; nested in Cases 3–4 at mean-field level; easy to simulate and to write BMVR ODEs.

Secondary: fixed fraction f if less burst-size noise than binomial is desired.

Later / data-driven: Beta random fraction, capped batch k, threshold rules, or short high-export windows — unless measurements demand them.

Takeaway. Between continuous budding and full dumping, let a release event remove a **distributed part** of the pool. The clean default is independent boolean release of each unit with probability q; with a load-dependent event clock this is Case 4 scaled by q, and with a constant clock it is mean-equivalent to Case 3 while still producing pulsed single-cell paths.

16 What is publishable, and how it fits together

16.1 Core claims suitable for a paper

1. **Closed-form single-cell theory** for the BDC, including simplified moments, corrected non-fixation probability, and a complete proof of the geometric burst-size law.
2. **Identity:** the burst-size distribution equals the quasi-stationary distribution (geometric with ratio $1/a$).
3. **Renewal BMVR** with closed-form kernels $\widehat{I}$ and δK, exact effective parameters $p_{\text{eff}}(r)$ and $d_{\mathcal{I},\text{eff}}(r)$, and invariant R_0.
4. **Growth-phase dependence of** p as a concrete, directional prediction for BMVR fitting practice.
5. **Identifiability:** three intracellular rates cannot be recovered from two BMVR parameters alone; three rescuing observables are named.
6. **Flooding advantage** (Result 8.1): the exact condition under which bursting beats matched-R_0 budding for early establishment.

16.2 Suggested paper structure

Target: *Journal of Theoretical Biology* or *Bulletin of Mathematical Biology*.

1. Introduction — BMVR's constant-release assumption; single-cell evidence; what is and is not new (renewal framework standard; closed-form kernels and consequences new).
2. The intracellular process — BDC, closed forms, QSD/burst-size identity.
3. Renewal reformulation — kernels and equations.
4. Exact reduction — p_{eff}, $d_{\mathcal{I},\text{eff}}$, characteristic equation, R_0 unchanged, p growth-phase dependent.
5. Identifiability — negative result and rescuing observables.
6. The flooding advantage — main theorem and the condition $1/\delta < 1/\beta + 1/\mu$.
7. Application — single-cell release data; lytic vs. budding caveat.
8. Discussion — limits of linear intracellular growth; two-type/eclipse extension as future work (or fold in if completed).

If the two-type kernels are finished, a natural split is: (i) one-type theory as above; (ii) two-type kernel with a mechanistic eclipse phase.

16.3 What this summary deliberately omits

- Full errata catalogue for thesis Chapters 1, 3–6 (recorded in the working notes).

- Open technical tasks (literature check, conceptual QSD proof, two-type moments, intermediate release models, nonlinear intracellular growth).
- The $N = 2$ immediate-transfer chain model (interesting, not yet at the same state of completion).
- Detailed hypergeometric expressions for the Laplace transforms (available in the notes; not needed for the main story).

17 Quick-reference formula sheet

Quantity	Closed form	Role
a, b	roots of the characteristic quadratic	structure
A, B	$a-1,\ 1-b$	simplify
$I(t)$	$(aB+bAw)/(B+Aw)$	no-burst survival
$D(t)$	$ab(w-1)/(aw-b)$	internal clearance
$\widehat{I}(t)$	$(a-b)^2w/[(B+Aw)(aw-b)]$	productive survival
$J(t)$	$(a-b)^2w/(B+Aw)^2$	mean load
$V(t)$	$(1-I)[1+(\beta/\delta)(1-I)]$	mean release
V_∞	$a(1-b)/(a-1)$	total mean output
$\varphi(t)$	$\delta J(t)$	burst-time density
$\mathbb{P}\{\mathcal{K}=k\}$	$(\delta/\beta)a^{-k}$	burst size
QSD	$(a-1)a^{-n}$	conditional load
$\mathbb{E}[T_{\text{prod}}]$	$\beta^{-1}\log(a/(a-1))$	productive lifetime
R_0	$\gamma T V_\infty/c$	threshold (unchanged)
$p_{\text{eff}}(r)$	$\delta\widetilde{K}/\widetilde{\widehat{I}}$	effective release
Flooding	$1/\delta < 1/\beta + 1/\mu$	when bursting wins
New viral dynamics (Result 5.1)		
$\mathcal{I}(t)$	$\mathcal{I}_0\widehat{I}(t) + (\gamma T\,\mathcal{V} * \widehat{I})(t)$	infected cells
$\mathcal{V}'(t)$	$\mathcal{I}_0 g(t) + (\gamma T\,\mathcal{V} * g)(t) - c\mathcal{V}$	free virions
$g(a)$	$\delta K(a) = V'(a)$	release kernel (BDC)
Reset catastrophe (Section 11)		
$S(a)$	$e^{-d_{\mathcal{I}}a}$	cell survival
$g(a)$	$\delta e^{-d_{\mathcal{I}}a}\mathbb{E}[X_a^2]$	multi-pulse release
BMVR	same convolutions as BDC	kernels only change
p_∞	$\delta\mathbb{E}_\pi[X^2]$	mature-cell limit
HIV coarse grain (Section 12)		
Intracellular mean	eclipse $\to m' \approx \nu - \mu m$	not $m' = rm$
Tat intermediate	rate $\leq$ late ON ceiling	no systematic overshoot

End of summary. Full derivations, numerical verification tables, and open tasks live in the BMVR extension working notes.